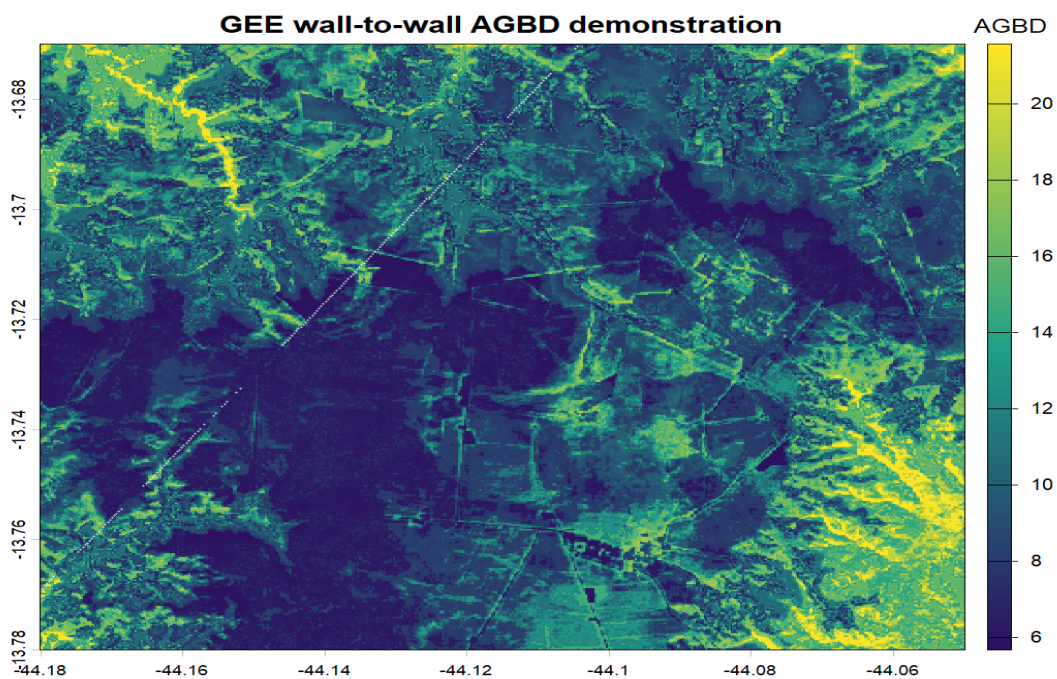


rGEDI

GEDI data access, processing, simulation, modeling, and mapping



Function reference • version 0.6.0 • 2026-10-09

114 documented help topics generated from the package source

Contents

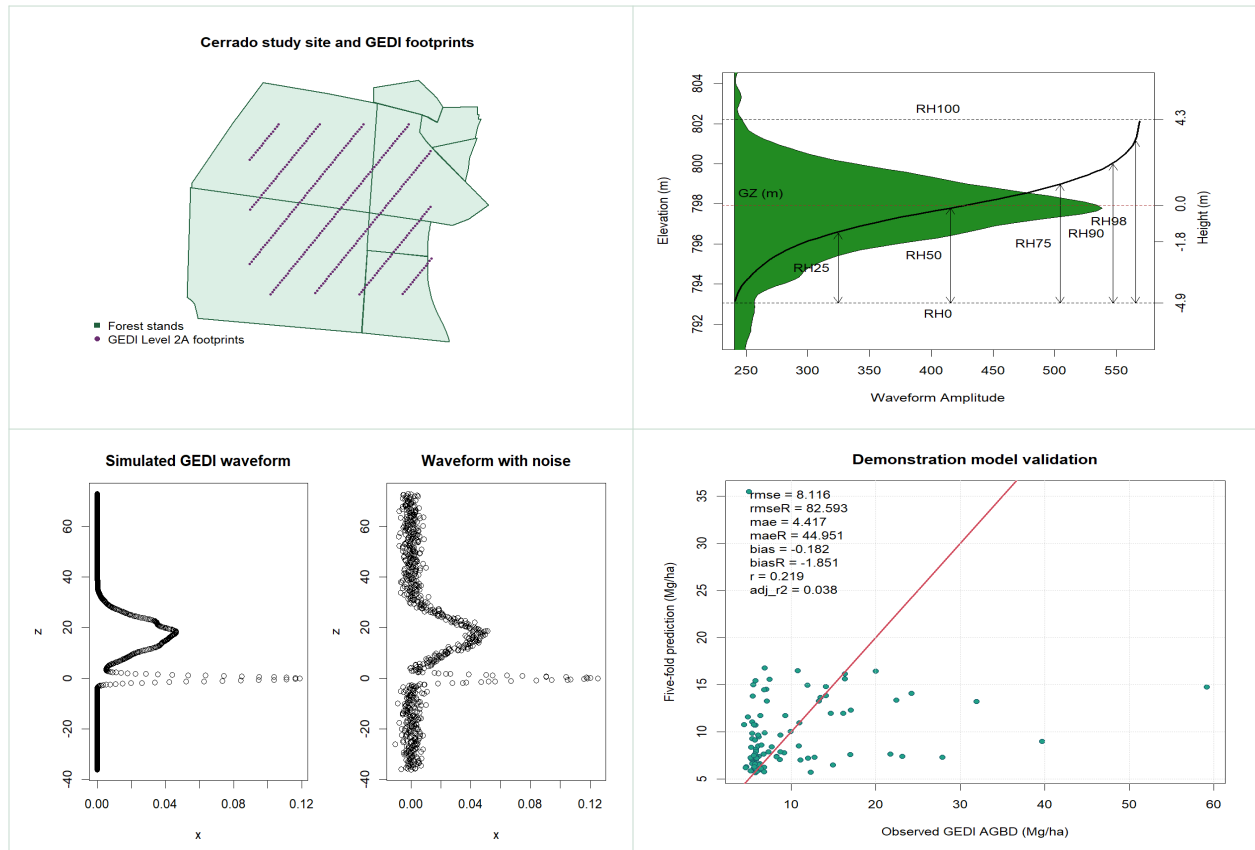
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Workflow at a glance

The README examples proceed from NASA CMR search and authenticated download through local or cloud reading, extraction, quality filtering, clipping, rasterization, modeling, simulation, and Earth Engine wall-to-wall mapping.



addEEImage

Description

Add an Earth Engine image to a Leaflet map

Usage

```
addEEImage(  
  map,  
  image,  
  bands = NULL,  
  min = 0,  
  max = 1,  
  palette = NULL,  
  group = NULL,  
  ...  
)
```

Arguments

map: A Leaflet widget.

image: Earth Engine image.

bands: One or three bands.

min, max: Visualization limits.

palette: Optional palette.

group: Layer group.

...: Passed to 'leaflet::addTiles()'.

Value

A Leaflet widget.

aggregateLevel3

Description

Aggregate a GEDI Level 3 raster

Usage

```
aggregateLevel3(x, fact, fun = mean, ...)
```

Arguments

x: A raster path or 'SpatRaster'.

fact: Aggregation factor.

fun: Aggregation function.

...: Passed to 'terra::aggregate()'.

Value

A 'terra::SpatRaster'.

aggregateLevel4B

Description

Aggregate a GEDI Level 4B raster

Usage

```
aggregateLevel4B(x, fact, fun = mean, ...)
```

Arguments

x: A raster path or 'SpatRaster'.

fact: Aggregation factor.

fun: Aggregation function.

...: Passed to 'terra::aggregate()'.

Value

A 'terra::SpatRaster'.

aspect

Description

Earth Engine terrain aspect

Usage

```
aspect(x)
```

Arguments

x: Earth Engine elevation image.

Value

An Earth Engine image.

build_ee_forest

Description

Train a regression forest in Earth Engine

Usage

```
build_ee_forest(training, response, predictors, trees = 500L, seed = 1L)
```

Arguments

training: Earth Engine 'FeatureCollection' containing predictors and the response.

response: Response property name, such as "agbd" or "rh98".

predictors: Predictor property names.

trees: Number of trees.

seed: Random seed.

Value

A trained Earth Engine classifier configured for regression.

clipLevel1B

Description

This function clips GEDI Level1B data (geolocated waveforms) within a given bounding coordinates

Usage

```
clipLevel1B(level1b, xmin, xmax, ymin, ymax, output)
```

Arguments

level1b: A 'gedi.level1b' object (output of 'readLevel1B()' function). An S4 object of class 'gedi.level1b'.

xmin: Numeric. West longitude (x) coordinate of the bounding rectangle, in decimal degrees.

xmax: Numeric. East longitude (x) coordinate of the bounding rectangle, in decimal degrees.

ymin: Numeric. South latitude (y) coordinate of the bounding rectangle, in decimal degrees.

ymax: Numeric. North latitude (y) coordinate of the bounding rectangle, in decimal degrees.

output: Optional character path where to save the new hdf5file. The default stores a temporary file only.

Value

Returns a list of S4 objects of class 'gedi.level1b' containing clipped GEDI Level1B data.

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi01-b-002>>

Examples

```
## Not run:

# Specifying the path to GEDI level1B data (zip file)
outdir <- tempdir()

level1B_fp_zip <- system.file("extdata",
  "GEDI01_B_2019108080338_001964_T05337_02_003_01_sub.zip",
  package = "rGEDI"
)

# Unzipping GEDI level1B data
level1Bpath <- unzip(level1B_fp_zip, exdir = outdir)

# Reading GEDI level1B data (h5 file)
level1b <- readLevel1B(level1Bpath = level1Bpath)

# Bounding rectangle coordinates
xmin <- -44.13
xmax <- -44.12
ymin <- -13.74
ymax <- -13.73

# Specifying output file and path
output <- file.path(outdir, "GEDI01_B_2019108080338_001964_T05337_02_003_01_clip")

# Clipping GEDI Level1B data by extent boundary box
level1b_clip <- clipLevel1B(level1b, xmin, xmax, ymin, ymax, output)

close(level1b)
close(level1b_clip)
## End(Not run)
```

clipLevel1BGeo

Description

This function clips GEDI level1B extracted geolocation ('getLevel1BGeo()') data a within given bounding coordinates

Usage

```
clipLevel1BGeo(level1BGeo, xmin, xmax, ymin, ymax)
```

Arguments

level1BGeo: A 'data.table::data.table' resulting from 'getLevel1BGeo()'.

xmin: Numeric. West longitude (x) coordinate of the bounding rectangle, in decimal degrees.

xmax: Numeric. East longitude (x) coordinate of the bounding rectangle, in decimal degrees.

ymin: Numeric. South latitude (y) coordinate of the bounding rectangle, in decimal degrees.

ymax: Numeric. North latitude (y) coordinate of the bounding rectangle, in decimal degrees.

Value

Returns an S4 object of class 'data.table::data.table'.

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi01-b-002>>

Examples

```
# Specifying the path to GEDI level1B data (zip file)
outdir <- tempdir()
level1B_fp_zip <- system.file("extdata",
  "GEDI01_B_2019108080338_001964_T05337_02_003_01_sub.zip",
  package = "rGEDI"
)

# Unzipping GEDI level1B data
level1Bpath <- unzip(level1B_fp_zip, exdir = outdir)

# Reading GEDI level1B data (h5 file)
level1b <- readLevel1B(level1Bpath = level1Bpath)

# Extracting GEDI Full Waveform Geolocations
level1bGeo <- getLevel1BGeo(level1b)

# Bounding rectangle coordinates
xmin <- -44.15036
xmax <- -44.10066
ymin <- -13.75831
ymax <- -13.71244

# Clipping GEDI Full Waveform Geolocations by boundary box extent
level1bGeo_clip <- clipLevel1BGeo(level1bGeo, xmin, xmax, ymin, ymax)

if (interactive() && requireNamespace("leaflet", quietly = TRUE)) {
  library(leaflet)
  leaflet() %>%
    addCircleMarkers(level1bGeo_clip$longitude_bin0,
      level1bGeo_clip$latitude_bin0,
      radius = 1,
      opacity = 1,
      color = "red"
    ) %>%
    addScaleBar(options = list(imperial = FALSE)) %>%
    addProviderTiles(providers$Esri.WorldImagery)
}

close(level1b)
```

clipLevel1BGeoGeometry

Description

This function clips level1BGeo extracted geolocation (level1BGeo) data within a given geometry

Usage

```
clipLevel1BGeoGeometry(level1BGeo, polygon, split_by = "id")
```

Arguments

level1BGeo: A 'data.table::data.table' resulting from 'getLevel1BGeo()' function.

polygon: Polygon. An object of class 'sf::sf', which can be loaded as an ESRI shapefile using sf::st_read function in the _sf_ package.

split_by: Polygon id. If defined, GEDI data will be clipped by each polygon using the polygon id from table of attribute defined by the user.

Value

Returns an S4 object of class 'data.table::data.table' containing the clipped GEDI level1B extracted geolocations.

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi01-b-002>>

Examples

```
# Specifying the path to GEDI level1B data (zip file)
outdir <- tempdir()
level1B_fp_zip <- system.file("extdata",
  "GEDI01_B_2019108080338_001964_T05337_02_003_01_sub.zip",
  package = "rGEDI"
)

# Unzipping GEDI level1B data
level1Bpath <- unzip(level1B_fp_zip, exdir = outdir)

# Reading GEDI level1B data (h5 file)
level1b <- readLevel1B(level1Bpath = level1Bpath)

# Extracting GEDI Full Waveform Geolocations
level1BGeo <- getLevel1BGeo(level1b, beams = "BEAM0001")

# Specifying the path to shapefile
polygon_filepath <- system.file("extdata", "stands_cerrado.shp", package = "rGEDI")

# Reading shapefile as sf object
library(sf)
polygon <- sf::st_read(polygon_filepath)

# Clipping GEDI Full Waveform Geolocations by Geometry
level1BGeo_clip <- clipLevel1BGeoGeometry(level1BGeo, polygon, split_by = "id")

if (interactive() && requireNamespace("leaflet", quietly = TRUE)) {
  library(leaflet)
  leaflet() %>%
    addCircleMarkers(level1BGeo_clip$longitude_bin0,
      level1BGeo_clip$latitude_bin0,
      radius = 1,
      opacity = 1,
      color = "red"
    ) %>%
    addScaleBar(options = list(imperial = FALSE)) %>%
    addPolygons(
      data = polygon, weight = 1, col = "white",
      opacity = 1, fillOpacity = 0
    ) %>%
    addProviderTiles(providers$Esri.WorldImagery)
}

close(level1b)
```

clipLevel1BGeometry

Description

This function clips GEDI Level1B (geolocated waveforms) data within a given bounding geometry

Usage

```
clipLevel1BGeometry(level1b, polygon, output = "", split_by = NULL)
```

Arguments

level1b: A 'gedi.level1b' object (output of 'readLevel1B()' function). An S4 object of class "gedi.level1b".

polygon: Polygon or Multipolygon. An object opened with 'sf::st_read',

output: Optional character path where to save the new 'hdf5::H5File'. The default stores a temporary file only.

split_by: Polygon id. If defined, GEDI data will be clipped by each polygon using the attribute specified by 'split_by' from the attribute table.

Value

Returns a list of S4 object of class 'gedi.level1b' containing clipped GEDI Level1B data.

Examples

```
## Not run:

outdir <- tempdir()

# Specifying the path to GEDI level1B data (zip file)
level1B_fp_zip <- system.file("extdata",
  "GEDI01_B_2019108080338_001964_T05337_02_003_01_sub.zip",
  package = "rGEDI"
)

# Unzipping GEDI level1B data
level1Bpath <- unzip(level1B_fp_zip, exdir = outdir)

# Reading GEDI level1B data (h5 file)
level1b <- readLevel1B(level1Bpath = level1Bpath)

# Specifying the path to shapefile
polygon_filepath <- system.file("extdata", "stands_cerrado.shp", package = "rGEDI")

# Reading shapefile as sf object
library(sf)
polygon <- sf::st_read(polygon_filepath)

# Specifying output file and path
output <- file.path(outdir, "GEDI01_B_2019108080338_001964_T05337_02_003_01_clip")

# clipping GEDI Level1B data by extent boundary box
level1b_clip <- clipLevel1BGeometry(level1b,
  polygon = polygon,
  output = output,
  split_by = "id"
)

close(level1b)
lapply(level1b_clip, close)
## End(Not run)
```

clipLevel2A

Description

This function clips GEDI Level2A data within a given bounding coordinates

Usage

```
clipLevel2A(level2a, xmin, xmax, ymin, ymax, output)
```

Arguments

level2a: A GEDI Level2A object (output of 'readLevel2A()' function). An S4 object of class "gedi.level2a".

xmin: Numeric. West longitude (x) coordinate of the bounding rectangle, in decimal degrees.

xmax: Numeric. East longitude (x) coordinate of the bounding rectangle, in decimal degrees.

ymin: Numeric. South latitude (y) coordinate of the bounding rectangle, in decimal degrees.

ymax: Numeric. North latitude (y) coordinate of the bounding rectangle, in decimal degrees.

output: Optional character path where to save the new hdf5file. The default stores a temporary file only.

Value

Returns a list of S4 objects of class "gedi.level2a" containing clipped GEDI Level2A data.

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi02-a-002>>

Examples

```
## Not run:

outdir <- tempdir()

# Specifying the path to GEDI level2A data (zip file)
level2A_fp_zip <- system.file("extdata",
  "GEDI02_A_2019108080338_001964_T05337_02_001_01_sub.zip",
  package = "rGEDI"
)

# Unzipping GEDI level2A data
level2Apath <- unzip(level2A_fp_zip, exdir = outdir)

# Reading GEDI level2A data (h5 file)
level2a <- readLevel2A(level2Apath = level2Apath)

# Bounding rectangle coordinates
xmin <- -44.13
xmax <- -44.12
ymin <- -13.74
ymax <- -13.73

print(level2a)

# Specifying output file and path
output <- file.path(outdir, "GEDI02_A_2019108080338_001964_T05337_02_001_01_clip.h5")

# Clipping GEDI Level2A data by boundary box extent
level2a_clip <- clipLevel2A(level2a, xmin, xmax, ymin, ymax, output)

close(level2a)
close(level2a_clip)
## End(Not run)
```

clipLevel2AGeometry

Description

This function clips GEDI Level2A data within a given geometry

Usage

```
clipLevel2AGeometry(level2a, polygon, output = "", split_by = NULL)
```

Arguments

level2a: A GEDI Level2A object (output of 'readLevel2A()' function). An S4 object of class "gedi.level2a".

polygon: Polygon. An object of class 'sf::sf', which can be loaded as an ESRI shapefile using 'sf::st_read()' function in the `_sf_` package.

output: optional character path where to save the new h5file. Default "" (temporary file).

split_by: Polygon id. If defined, GEDI data will be clipped by each polygon using the attribute specified by 'split_by' from the attribute table.

Value

Returns a list of S4 object of class "gedi.level2a" containing clipped GEDI Level2A data.

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi02-a-002>>

Examples

```
## Not run:

outdir <- tempdir()

# Specifying the path to GEDI level2A data (zip file)
level2A_fp_zip <- system.file("extdata",
  "GEDI02_A_2019108080338_001964_T05337_02_001_01_sub.zip",
  package = "rGEDI"
)

# Unzipping GEDI level2A data
level2Apath <- unzip(level2A_fp_zip, exdir = outdir)

# Reading GEDI level2A data (h5 file)
level2a <- readLevel2A(level2Apath = level2Apath)

# Specifying the path to shapefile
polygon_filepath <- system.file("extdata", "stands_cerrado.shp", package = "rGEDI")

# Reading shapefile as sf object
library(sf)
polygon <- sf::st_read(polygon_filepath)

# Specifying output file and path
output <- file.path(outdir, "GEDI02_A_2019108080338_001964_T05337_02_001_01_clip")

# Clipping GEDI Level2A data by geometry
level2a_clip <- clipLevel2AGeometry(level2a,
  polygon = polygon,
  output = output,
  split_by = "id"
)
close(level2a)
lapply(level2a_clip, close)
## End(Not run)
```


clipLevel2AM

Description

This function clips GEDI Level2A extracted Elevation and Height Metrics (Level2AM) within a given bounding coordinates

Usage

```
clipLevel2AM(level2AM, xmin, xmax, ymin, ymax)
```

Arguments

level2AM: A GEDI Level2A object (output of 'readLevel2A()' function). An S4 object of class "gedi.level2a".

xmin: Numeric. West longitude (x) coordinate of bounding rectangle, in decimal degrees.

xmax: Numeric. East longitude (x) coordinate of bounding rectangle, in decimal degrees.

ymin: Numeric. South latitude (y) coordinate of bounding rectangle, in decimal degrees.

ymax: Numeric. North latitude (y) coordinate of bounding rectangle, in decimal degrees.

Value

Returns an S4 object of class data.table::data.table containing the clipped elevation and relative heights metrics.

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi02-a-002>>

Examples

```
# Specifying the path to GEDI level2A data (zip file)
outdir <- tempdir()
level2A_fp_zip <- system.file("extdata",
  "GEDI02_A_2019108080338_001964_T05337_02_001_01_sub.zip",
  package = "rGEDI"
)

# Unzipping GEDI level2A data
level2Apath <- unzip(level2A_fp_zip, exdir = outdir)

# Reading GEDI level2A data (h5 file)
level2a <- readLevel2A(level2Apath = level2Apath)

# Extracting GEDI Elevation and Height Metrics
level2AM <- getLevel2AM(level2a, beams = "BEAM0001")

# Bounding rectangle coordinates
xmin <- -44.15036
xmax <- -44.10066
ymin <- -13.75831
ymax <- -13.71244

# Clipping GEDI data by boundary box extent
level2AM_clip <- clipLevel2AM(level2AM, xmin, xmax, ymin, ymax)

close(level2a)
```

clipLevel2AMGeometry

Description

This function clips GEDI Level2A extracted Elevation and Height Metrics (Level2AM) within a given bounding coordinates

Usage

```
clipLevel2AMGeometry(level2AM, polygon, split_by = "id")
```

Arguments

level2AM: A GEDI Level2A object (output of 'readLevel2A()' function). An S4 object of class "data.table".

polygon: Polygon. An object of class 'sf::sf', which can be loaded as an ESRI shapefile using sf::st_read function in the _sf_ package.

split_by: Polygon id. If defined, GEDI data will be clipped by each polygon using the polygon id from table of attribute defined by the user

Value

Returns an S4 object of class data.table::data.table containing the clipped elevation and relative heights metrics.

Examples

```
# Specifying the path to GEDI level2A data (zip file)
outdir <- tempdir()
level2A_fp_zip <- system.file("extdata",
  "GEDI02_A_2019108080338_001964_T05337_02_001_01_sub.zip",
  package = "rGEDI"
)

# Unzipping GEDI level2A data
level2Apath <- unzip(level2A_fp_zip, exdir = outdir)

# Reading GEDI level2A data (h5 file)
level2a <- readLevel2A(level2Apath = level2Apath)

# Extracting GEDI Elevation and Height Metrics
level2AM <- getLevel2AM(level2a, beams = "BEAM0001")

# Specifying the path to shapefile
polygon_filepath <- system.file("extdata", "stands_cerrado.shp", package = "rGEDI")

# Reading shapefile as sf object
library(sf)
polygon <- sf::st_read(polygon_filepath)

# Clipping GEDI data by Geometry
level2AM_clip <- clipLevel2AMGeometry(level2AM, polygon, split_by = "id")

hasLeaflet <- require(leaflet)

if (hasLeaflet) {
  leaflet() %>%
    addCircleMarkers(level2AM_clip$lat_lowestmode,
      level2AM_clip$lon_lowestmode,
      radius = 1,
      opacity = 1,
      color = "red"
    ) %>%
    addScaleBar(options = list(imperial = FALSE)) %>%
    addPolygons(
      data = polygon, weight = 1, col = "white",
      opacity = 1, fillOpacity = 0
    ) %>%
    addProviderTiles(providers$Esri.WorldImagery)
}

close(level2a)
```

clipLevel2B

Description

This function extracts GEDI Level1B data a within given bounding coordinates

Usage

```
clipLevel2B(level2b, xmin, xmax, ymin, ymax, output = "")
```

Arguments

level2b: A GEDI Level2B object (output of 'readLevel2B()' function). An S4 object of class "gedi.level2b".

xmin: Numeric. West longitude (x) coordinate of the bounding rectangle, in decimal degrees.

xmax: Numeric. East longitude (x) coordinate of the bounding rectangle, in decimal degrees.

ymin: Numeric. South latitude (y) coordinate of the bounding rectangle, in decimal degrees.

ymax: Numeric. North latitude (y) coordinate of the bounding rectangle, in decimal degrees.

output: Optional character path where to save the new hdf5 file. The default stores a temporary file only.

Value

Returns a list of S4 object of class "gedi.level2b" containing clipped GEDI Level2B data.

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi01-b-002>>

Examples

```
## Not run:

outdir <- tempdir()

# Specifying the path to GEDI level2B data (zip file)
level2B_fp_zip <- system.file("extdata",
  "GEDI02_B_2019108080338_001964_T05337_02_001_01_sub.zip",
  package = "rGEDI"
)

# Unzipping GEDI level2A data
level2Bpath <- unzip(level2B_fp_zip, exdir = outdir)

# Reading GEDI level2B data (h5 file)
level2b <- readLevel2B(level2Bpath = level2Bpath)

# Bounding rectangle coordinates
xmin <- -44.13
xmax <- -44.12
ymin <- -13.74
ymax <- -13.73

# Specifying output file and path
output <- file.path(outdir, "GEDI02_B_2019108080338_001964_T05337_02_001_01_clip")

# Clipping GEDI data by extent boundary box
level2b_clip <- clipLevel2B(level2b, xmin, xmax, ymin, ymax)

close(level2b)
close(level2b_clip)
## End(Not run)
```

clipLevel2BGeometry

Description

This function extracts GEDI Level1B data within a given geometry

Usage

```
clipLevel2BGeometry(level2b, polygon, output = "", split_by = NULL)
```

Arguments

level2b: A GEDI Level2B object (output of 'readLevel2B()' function). An S4 object of class "gedi.level2b".

polygon: Polygon. An object of class 'sf::sf', which can be loaded as an ESRI shapefile using sf::st_read function in the _sf_ package.

output: optional character path where to save the new h5file. Default "" (temporary file).

split_by: Polygon id. If defined, GEDI data will be clipped by each polygon using the attribute specified by 'split_by' from the attribute table.

Value

Returns a list of S4 objects of class "gedi.level2b" containing clipped GEDI Level2B data.

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi01-b-002>>

Examples

```
## Not run:

outdir <- tempdir()

# Specifying the path to GEDI level2B data (zip file)
level2B_fp_zip <- system.file("extdata",
  "GEDI02_B_2019108080338_001964_T05337_02_001_01_sub.zip",
  package = "rGEDI"
)

# Unzipping GEDI level2A data
level2Bpath <- unzip(level2B_fp_zip, exdir = outdir)

# Reading GEDI level2B data (h5 file)
level2b <- readLevel2B(level2Bpath = level2Bpath)

# Specifying the path to shapefile
polygon_filepath <- system.file("extdata", "stands_cerrado.shp", package = "rGEDI")

# Reading shapefile as sf object
library(sf)
polygon <- sf::st_read(polygon_filepath)

# Specifying output file and path
output <- file.path(outdir, "GEDI02_B_2019108080338_001964_T05337_02_001_01_clip")

# Clipping GEDI data by extent boundary box
level2b_clip <- clipLevel2BGeometry(level2b,
  polygon = polygon,
  output = output,
  split_by = "id"
)

close(level2b)
lapply(level2b_clip, close)
## End(Not run)
```

clipLevel2BPAIProfile

Description

This function clips GEDI level2B derived Plant Area Index profile a within given bounding coordinates

Usage

```
clipLevel2BPAIProfile(level2BPAIProfile, xmin, xmax, ymin, ymax)
```

Arguments

level2BPAIProfile: A GEDI Level2B object (output of 'getLevel2BPAIProfile()' function). An S4 object of class "gedi.level2b".

xmin: Numeric. West longitude (x) coordinate of the bounding rectangle, in decimal degrees.

xmax: Numeric. East longitude (x) coordinate of the bounding rectangle, in decimal degrees.

ymin: Numeric. South latitude (y) coordinate of the bounding rectangle, in decimal degrees.

ymax: Numeric. North latitude (y) coordinate of the bounding rectangle, in decimal degrees.

Value

Returns an S4 object of class data.table::data.table containing the Plant Area Index profile data.

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi02-b-002>>

Examples

```
# Specifying the path to GEDI level2B data (zip file)
outdir <- tempdir()
level2B_fp_zip <- system.file("extdata",
  "GEDI02_B_2019108080338_001964_T05337_02_001_01_sub.zip",
  package = "rGEDI"
)

# Unzipping GEDI level2A data
level2Bpath <- unzip(level2B_fp_zip, exdir = outdir)

# Reading GEDI level2B data (h5 file)
level2b <- readLevel2B(level2Bpath = level2Bpath)

# Extracting GEDI Plant Area Index profile
level2BPAIProfile <- getLevel2BPAIProfile(level2b)

# Bounding rectangle coordinates
xmin <- -44.15036
xmax <- -44.10066
ymin <- -13.75831
ymax <- -13.71244

# Clipping GEDI Plant Area Index profile by extent boundary box
level2b_clip <- clipLevel2BPAIProfile(level2BPAIProfile, xmin, xmax, ymin, ymax)

close(level2b)
```

clipLevel2BPAlProfileGeometry

Description

This function clips GEDI level2B derived Plant Area Index profile within a given geometry

Usage

```
clipLevel2BPAlProfileGeometry(level2BPAlProfile, polygon, split_by = NULL)
```

Arguments

level2BPAlProfile: A GEDI Level2B object (output of 'getLevel2BPAlProfile()' function). An S4 object of class "data.table".

polygon: Polygon. An object of class 'sf::sf', which can be loaded as an ESRI shapefile using `sf::st_read` function in the `_sf_` package.

split_by: Polygon id. If defined, GEDI data will be clipped by each polygon using the attribute specified by 'split_by' from the attribute table.

Value

Returns an S4 object of class `data.table::data.table` containing the Plant Area Index profile data.

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi02-b-002>>

Examples

```
# Specifying the path to GEDI level2B data (zip file)
outdir <- tempdir()
level2B_fp_zip <- system.file("extdata",
  "GEDI02_B_2019108080338_001964_T05337_02_001_01_sub.zip",
  package = "rGEDI"
)

# Unzipping GEDI level2A data
level2Bpath <- unzip(level2B_fp_zip, exdir = outdir)

# Reading GEDI level2B data (h5 file)
level2b <- readLevel2B(level2Bpath = level2Bpath)

# Extracting GEDI Plant Area Index profile
level2BPAlProfile <- getLevel2BPAlProfile(level2b)

# Specifying the path to shapefile
polygon_filepath <- system.file("extdata", "stands_cerrado.shp", package = "rGEDI")

# Reading shapefile as sf object
library(sf)
polygon <- sf::st_read(polygon_filepath)

# Clipping GEDI Plant Area Index profile by geometry
level2b_clip_geometry <- clipLevel2BPAlProfileGeometry(
  level2BPAlProfile,
  polygon,
  split_by = "id"
)

close(level2b)
```

clipLevel2BPAVDProfile

Description

This function clips GEDI level2B derived Plant Area Volume Density profile within a given bounding coordinates

Usage

```
clipLevel2BPAVDProfile(level2BPAVDProfile, xmin, xmax, ymin, ymax)
```

Arguments

level2BPAVDProfile: A GEDI Level2B object (output of 'getLevel2BPAVDProfile()' function). An S4 object of class "data.table".

xmin: Numeric. West longitude (x) coordinate of the bounding rectangle, in decimal degrees.

xmax: Numeric. East longitude (x) coordinate of the bounding rectangle, in decimal degrees.

ymin: Numeric. South latitude (y) coordinate of the bounding rectangle, in decimal degrees.

ymax: Numeric. North latitude (y) coordinate of the bounding rectangle, in decimal degrees.

Value

Returns an S4 object of class data.table::data.table containing the Plant Area Volume Density profile data.

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi02-b-002>>

Examples

```
# specify the path to GEDI level2B data (zip file)
outdir <- tempdir()
level2B_fp_zip <- system.file("extdata",
  "GEDI02_B_2019108080338_001964_T05337_02_001_01_sub.zip",
  package = "rGEDI"
)

# Unzipping GEDI level2A data
level2Bpath <- unzip(level2B_fp_zip, exdir = outdir)

# Reading GEDI level2B data (h5 file)
level2b <- readLevel2B(level2Bpath = level2Bpath)

# Extracting GEDI Plant Area Volume Density profile
level2BPAVDProfile <- getLevel2BPAVDProfile(level2b)

# Bounding rectangle coordinates
xmin <- -44.15036
xmax <- -44.10066
ymin <- -13.75831
ymax <- -13.71244

# Clipping GEDI Plant Area Volume Density profile by boundary box extent
level2BPAVDProfile_clip <- clipLevel2BPAVDProfile(level2BPAVDProfile, xmin, xmax, ymin, ymax)

close(level2b)
```

clipLevel2BPAVDProfileGeometry

Description

This function clips GEDI level2B derived Plant Area Index profile within a given geometry

Usage

```
clipLevel2BPAVDProfileGeometry(level2BPAVDProfile, polygon, split_by = NULL)
```

Arguments

level2BPAVDProfile: A GEDI Level2B object (output of 'getLevel2BPAIProfile()' function). An S4 object of class "gedi.level2b".

polygon: Polygon. An object of class 'sf::sf', which can be loaded as an ESRI shapefile using `sf::st_read` function in the `_sf_` package.

split_by: Polygon id. If defined, GEDI data will be clipped by each polygon using the attribute specified by 'split_by' from the attribute table.

Value

Returns an S4 object of class `data.table::data.table` containing the Plant Area Volume Density profile data.

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi02-b-002>>

Examples

```
# Specifying the path to GEDI level2B data (zip file)
outdir <- tempdir()
level2B_fp_zip <- system.file("extdata",
  "GEDI02_B_2019108080338_001964_T05337_02_001_01_sub.zip",
  package = "rGEDI"
)

# Unzipping GEDI level2A data
level2Bpath <- unzip(level2B_fp_zip, exdir = outdir)

# Reading GEDI level2B data (h5 file)
level2b <- readLevel2B(level2Bpath = level2Bpath)

# Extracting GEDI Plant Area Volume Density profile
level2BPAVDProfile <- getLevel2BPAVDProfile(level2b)

# Specifying the path to shapefile
polygon_filepath <- system.file("extdata", "stands_cerrado.shp", package = "rGEDI")

# Reading shapefile as sf object
library(sf)
polygon <- sf::st_read(polygon_filepath)

# Clipping GEDI Plant Area Volume Density profile by geometry
level2BPAVDProfile_clip <- clipLevel2BPAVDProfileGeometry(
  level2BPAVDProfile,
  polygon,
  split_by = "id"
)

close(level2b)
```


clipLevel2BVPM

Description

This function clips GEDI level2B derived Canopy Cover and Vertical Profile metrics a within given bounding coordinates

Usage

```
clipLevel2BVPM(level2BVPM, xmin, xmax, ymin, ymax)
```

Arguments

level2BVPM: A GEDI Level2B object (output of 'readLevel1B()' function). An S4 object of class "data.table".

xmin: Numeric. West longitude (x) coordinate of the bounding rectangle, in decimal degrees.

xmax: Numeric. East longitude (x) coordinate of the bounding rectangle, in decimal degrees.

ymin: Numeric. South latitude (y) coordinate of the bounding rectangle, in decimal degrees.

ymax: Numeric. North latitude (y) coordinate of the bounding rectangle, in decimal degrees.

Value

Returns an S4 object of class data.table::data.table containing the Canopy Cover and Vertical Profile metrics.

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi02-b-002>>

Examples

```
# Specifying the path to GEDI level2B data (zip file)
outdir <- tempdir()
level2B_fp_zip <- system.file("extdata",
  "GEDI02_B_2019108080338_001964_T05337_02_001_01_sub.zip",
  package = "rGEDI"
)

# Unzipping GEDI level2A data
level2Bpath <- unzip(level2B_fp_zip, exdir = outdir)

# Reading GEDI level2B data (h5 file)
level2b <- readLevel2B(level2Bpath = level2Bpath)

# Extracting canopy cover and vertical profile metrics
beam <- grep("^BEAM", (level2b@h5)$ls(recursive = FALSE)$name,
  value = TRUE)[1]
level2BVPM <- getLevel2BVPM(level2b, beams = beam)

# Bounding rectangle coordinates
xmin <- -44.15036
xmax <- -44.10066
ymin <- -13.75831
ymax <- -13.71244

# Clipping level2BVPM by extent boundary box
level2b_clip <- clipLevel2BVPM(level2BVPM, xmin, xmax, ymin, ymax)

hasLeaflet <- require(leaflet)

if (hasLeaflet) {
  leaflet() %>%
    addCircleMarkers(level2b_clip$longitude_bin0,
      level2b_clip$latitude_bin0,
      radius = 1,
      opacity = 1,
      color = "red"
    ) %>%
    addScaleBar(options = list(imperial = FALSE)) %>%
    addProviderTiles(providers$Esri.WorldImagery)
}

close(level2b)
```

clipLevel2BVPMGeometry

Description

This function clips GEDI level2B derived Canopy Cover and Vertical Profile metrics within a given geometry

Usage

```
clipLevel2BVPMGeometry(level2BVPM, polygon, split_by = NULL)
```

Arguments

level2BVPM: A GEDI Level2B object (output of 'readLevel1B()' function). An S4 object of class "gedi.level2b".

polygon: Polygon. An object of class 'sf::sf', which can be loaded as an ESRI shapefile using `sf::st_read` function in the `_sf_` package.

split_by: Polygon id. If defined, GEDI data will be clipped by each polygon using the attribute specified by 'split_by' from the attribute table.

Value

Returns an S4 object of class `data.table::data.table` containing the Canopy Cover and Vertical Profile metrics.

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi02-b-002>>

Examples

```
# Specifying the path to GEDI level2B data (zip file)
outdir <- tempdir()
level2B_fp_zip <- system.file("extdata",
  "GEDI02_B_2019108080338_001964_T05337_02_001_01_sub.zip",
  package = "rGEDI"
)

# Unzipping GEDI level2A data
level2Bpath <- unzip(level2B_fp_zip, exdir = outdir)

# Reading GEDI level2B data (h5 file)
level2b <- readLevel2B(level2Bpath = level2Bpath)

# Extracting canopy cover and vertical profile metrics
beam <- grep("^BEAM", (level2b@h5)$ls(recursive = FALSE)$name,
  value = TRUE)[1]
level2BVPM <- getLevel2BVPM(level2b, beams = beam)

# Specifying the path to shapefile
polygon_filepath <- system.file("extdata", "stands_cerrado.shp", package = "rGEDI")

# Reading shapefile as sf object
library(sf)
polygon <- sf::st_read(polygon_filepath)

# Clipping level2BVPM by geometry
level2b_clip_geometry <- clipLevel2BVPMGeometry(level2BVPM, polygon, split_by = "id")

hasLeaflet <- require(leaflet)

if (hasLeaflet) {
  leaflet() %>%
    addCircleMarkers(level2b_clip_geometry$longitude_bin0,
      level2b_clip_geometry$latitude_bin0,
      radius = 1,
      opacity = 1,
      color = "red"
    ) %>%
    addScaleBar(options = list(imperial = FALSE)) %>%
    addPolygons(
      data = polygon, weight = 1, col = "white",
      opacity = 1, fillOpacity = 0
    ) %>%
    addProviderTiles(providers$Esri.WorldImagery)
}

close(level2b)
```

clipLevel3

Description

Clip a GEDI Level 3 raster

Usage

```
clipLevel3(x, y, filename = "", overwrite = FALSE)
```

Arguments

x: A raster path or 'SpatRaster'.

y: An extent 'c(xmin, xmax, ymin, ymax)' or polygon object.

filename: Optional output filename.

overwrite: Logical.

Value

A 'terra::SpatRaster'.

clipLevel4A

Description

Clip GEDI Level 4A footprints by an extent

Usage

```
clipLevel4A(level4A, xmin, xmax, ymin, ymax)
```

Arguments

level4A: A table returned by 'getLevel4A()'.

xmin, xmax, ymin, ymax: Bounding coordinates in longitude/latitude.

Value

A `data.table::data.table`.

clipLevel4AGeometry

Description

Clip GEDI Level 4A footprints by polygons

Usage

```
clipLevel4AGeometry(level4A, polygon, split_by = NULL)
```

Arguments

level4A: A table returned by 'getLevel4A()'.

polygon: An 'sf', 'sfc', or 'SpatVector' polygon object.

split_by: Optional polygon attribute copied to 'poly_id'.

Value

A data.table::data.table.

clipLevel4B

Description

Clip a GEDI Level 4B raster

Usage

```
clipLevel4B(x, y, filename = "", overwrite = FALSE)
```

Arguments

x: A raster path or 'SpatRaster'.

y: An extent 'c(xmin, xmax, ymin, ymax)' or polygon object.

filename: Optional output filename.

overwrite: Logical.

Value

A 'terra::SpatRaster'.

close

Description

Closing files will avoid locking HDF5 GEDI files.

Closing files will avoid locking HDF5 GEDI files.

Closing files will avoid locking HDF5 GEDI files.

Usage

```
## S4 method for signature 'gedi.level1b'
close(con, ...)

## S4 method for signature 'gedi.level2a'
close(con, ...)

## S4 method for signature 'gedi.level2b'
close(con, ...)

## S4 method for signature 'gedi.level4a'
close(con, ...)
```

Arguments

con: An object of class 'gedi.level2b'

...: Inherited from base

close.gedi.cloud_h5

Description

Close a cloud HDF5 connection

Usage

```
## S3 method for class 'gedi.cloud_h5'  
close(con, ...)
```

Arguments

con: A 'gedi.cloud_h5' object returned by 'readGEDICloud()'.
...: Reserved for compatibility with 'close()'.

Value

Invisibly returns 'NULL'.

earthdata_login

Description

Uses an existing bearer token, environment credentials, or '.netrc' file.

Usage

```
earthdata_login(persist = TRUE, netrc = Sys.getenv("NETRC", unset = ""))
```

Arguments

persist: Logical; allow Earthaccess to persist an interactive login.

netrc: Path to an Earthdata '.netrc' file.

Value

Invisibly returns 'TRUE' after successful authentication.

earthdata_s3_credentials

Description

Requests short-lived AWS credentials from Earthaccess and exports them for GDAL/terra. Direct S3 access to GEDI is available from AWS 'us-west-2'.

Usage

```
earthdata_s3_credentials(daac = "ORNL")
```

Arguments

daac: NASA DAAC short name. GEDI products are hosted by ORNL DAAC.

Value

Invisibly returns the credential list.

ee_build_AlphaEarth_embedding_terrain_stack

Description

Build an AlphaEarth and terrain predictor stack

Usage

```
ee_build_AlphaEarth_embedding_terrain_stack(  
  geom,  
  start_year,  
  end_year,  
  add_lonlat = TRUE  
)
```

Arguments

geom: Area of interest.

start_year, end_year: Inclusive year range.

add_lonlat: Add coordinate bands.

Value

An Earth Engine image.

ee_build_hls_s1c_terrain_stack

Description

Build an HLS and terrain predictor stack in Earth Engine

Usage

```
ee_build_hls_s1c_terrain_stack(  
  x,  
  start_date,  
  end_date,  
  cloud_max = 20,  
  buffer_m = 30  
)
```

Arguments

x: Area of interest.

start_date, end_date: Date range.

cloud_max: Maximum HLS cloud cover.

buffer_m: AOI buffer in meters.

Value

An Earth Engine image.

ee_check_task_status

Description

Check an Earth Engine task

Usage

```
ee_check_task_status(task, quiet = TRUE)
```

Arguments

task: Earth Engine batch task.

quiet: Logical.

Value

Task status list.

ee_image_to_drive

Description

Creates a Google Earth Engine batch export task for a GeoTIFF in Google Drive. This is useful for images that exceed the direct download limit.

Usage

```
ee_image_to_drive(  
    image,  
    description = "rGEDI_export",  
    folder = "EE_Exports",  
    file_name_prefix = description,  
    region,  
    scale = 30,  
    crs = "EPSG:4326",  
    max_pixels = 1e+13,  
    start = TRUE,  
    ...  
)
```

Arguments

image: Earth Engine image.

description: Task description shown in Earth Engine.

folder: Google Drive folder.

file_name_prefix: Output file prefix.

region: Area of interest accepted by 'ext_to_ee()'.

scale: Pixel size in meters.

crs: Output CRS.

max_pixels: Maximum number of pixels allowed by Earth Engine.

start: Start the task immediately.

...: Additional arguments passed to Earth Engine's 'toDrive' method.

Value

An Earth Engine batch task. Use 'ee_check_task_status()' to monitor it.

ee_initialize

Description

Initialize Google Earth Engine

Usage

```
ee_initialize(  
    project = Sys.getenv("EE_PROJECT", unset = ""),  
    authenticate = TRUE,  
    quiet = FALSE  
)
```

Arguments

project: Google Cloud project ID.

authenticate: Authenticate when initialization fails.

quiet: Logical.

Value

Invisibly returns the Python Earth Engine module.

ee_rect_to_sf

Description

Convert an Earth Engine rectangle to 'sf'

Usage

```
ee_rect_to_sf(aoi)
```

Arguments

aoi: Earth Engine geometry.

Value

An 'sf' polygon.

ext_to_ee

Description

Convert an extent to an Earth Engine geometry

Usage

```
ext_to_ee(x)
```

Arguments

x: A 'SpatExtent', raster/vector object, or numeric extent in 'c(xmin, xmax, ymin, ymax)' order.

Value

An Earth Engine geometry.

extractEE

Description

Extract Earth Engine image values at GEDI footprints

Usage

```
extractEE(stack, geom, scale = 30, chunk_size = 1000L)

extractGEDIAncillary(stack, geom, scale = 30, chunk_size = 1000L)
```

Arguments

stack: Earth Engine image or list of images.

geom: GEDI table, 'sf', or 'SpatVector' points.

scale: Sampling scale in meters.

chunk_size: Number of points per request.

Value

A data.table::data.table.

extractLevel3

Description

Extract GEDI Level 3 raster values

Usage

```
extractLevel3(x, y, ...)
```

Arguments

x: A raster path or 'SpatRaster'.

y: Points or polygons accepted by 'terra::vect()'.

...: Passed to 'terra::extract()'.

Value

A data frame.

extractLevel4B

Description

Extract GEDI Level 4B raster values

Usage

```
extractLevel4B(x, y, ...)
```

Arguments

x: A raster path or 'SpatRaster'.

y: Points or polygons accepted by 'terra::vect()'.

...: Passed to 'terra::extract()'.

Value

A data frame.

fit_metrics

Description

Model fit metrics

Usage

```
fit_metrics(observed, predicted, plotstat = FALSE, unit = "", ...)
```

Arguments

observed, predicted: Numeric vectors.

plotstat: Draw an observed-versus-predicted plot.

unit: Unit label.

...: Passed to 'graphics::plot()'.

Value

A data frame of accuracy statistics.

fit_model

Description

Fit a GEDI statistical or random-forest model

Usage

```
fit_model(  
  x,  
  y,  
  method = c("randomForest", "lm"),  
  test = c("none", "split", "kfold", "loocv", "bootstrap"),  
  k = 5L,  
  test_size = 0.3,  
  iterations = 100L,  
  seed = NULL,  
  ...  
)
```

Arguments

x: Predictor data frame.

y: Response vector or response column name in 'x'.

method: "randomForest" or "lm".

test: Validation method: "none", "split", "kfold", "loocv", or "bootstrap".

k: Number of folds.

test_size: Fraction held out for split validation.

iterations: Bootstrap iterations.

seed: Optional random seed.

...: Model arguments.

Value

An object of class 'gedi_model'.

gedi.fullwaveform-class

Description

Class for GEDI level1B Full Waveform

Slots:

'dt' Object of class data.table from `_data.table_` package containing the extracted GEDI full-waveform elevation and amplitude.

gedi.level1b-class

Description

Class for GEDI level1B

Slots:

'h5' A local 'H5File' or a cloud-backed GEDI HDF5 connection containing geolocated waveforms.

See Also

'H5File' in the 'hdf5r' package and <<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi01-b-002>>

gedi.level2a-class

Description

Class for GEDI level2A

Slots:

'h5' A local 'H5File' or cloud-backed GEDI HDF5 connection containing ground elevation, canopy top height, and relative heights (RH).

See Also

'H5File' in the 'hdf5r' package and <<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi02-a-002>>

gedi.level2b-class

Description

Class for GEDI level2B

Slots:

'h5' A local 'H5File' or cloud-backed GEDI HDF5 connection containing canopy cover, PAI, PAVD, and FHD.

See Also

'H5File' in the 'hdf5r' package and <<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi02-b-002>>

gedi.level4a-class

Description

Class for GEDI Level 4A footprint biomass data

Slots:

'h5' An open local or cloud-backed HDF5 connection containing a GEDI04_A granule.

See Also

<https://daac.ornl.gov/GEDI/guides/GEDI_L4A_AGB_Density_V3.html>

gediDownload

Description

Download GEDI data from LP DAAC Data Pool. Users will need to enter their Earth Explore login Information for downloading the data.

Usage

```
gediDownload(  
  filepath,  
  outdir = NULL,  
  overwrite = FALSE,  
  buffer_size = 512,  
  timeout = 10,  
  netrc = Sys.getenv("NETRC", unset = "")  
)  
  
downloadGEDI(  
  filepath,  
  outdir = NULL,  
  overwrite = FALSE,  
  buffer_size = 512,  
  timeout = 10,  
  netrc = Sys.getenv("NETRC", unset = "")  
)
```

Arguments

filepath: Vector object; path to the GEDI data

outdir: Vector object, output directory for downloading GEDI data, default 'tempdir()'

overwrite: logical; overwrite file if they already exists in destination, default FALSE

buffer_size: integer; the size of download chunk in KB to hold in memory before writing to file, default 512.

timeout: integer; connection timeout in seconds.

netrc: Optional path to an existing NASA Earthdata '.netrc' file. By default, uses the 'NETRC' environment variable before prompting.

Value

No return value on success, on failure it will 'stop()'

References

Credits to Cole Krehbiel. Code adapted from

<https://git.earthdata.nasa.gov/projects/LPDUR/repos/daac_data_download_r/browse/DAACDataDownload.R>

Examples

```
## Not run:  
  
# Set path to GEDI data  
# herein we will only download xml metedata  
filepath=c(paste0(  
  "https://e4ftl01.cr.usgs.gov/GEDI/GEDI02_B.001",  
  "/2019.04.18/GEDI02_B_2019108032534_001961_T03911_02_001_01.h5.xml"  
),  
  paste0("https://e4ftl01.cr.usgs.gov/GEDI/GEDI02_B.001",  
    "/2019.04.18/GEDI02_B_2019108045815_001962_T01066_02_001_01.h5.xml"  
  )  
)  
  
# Set dir to download files to  
outdir=tempdir()  
  
# Create .netrc file  
netrc = file.path(outdir, ".netrc")  
netrc_conn <- file(netrc)  
  
writeLines(c("machine urs.earthdata.nasa.gov",  
  sprintf("login %s", Sys.getenv("NASA_USER")),  
  sprintf("password %s", Sys.getenv("NASA_PASSWORD"))  
, netrc_conn)  
  
close(netrc_conn)  
#' Downloading GEDI data
```

```
gediDownload(filepath,outdir)
## End(Not run)
```

gediEE

Description

Open a GEDI product in Google Earth Engine

Usage

```
gediEE(  
  product = names(.gedi_ee_catalog),  
  start_date = NULL,  
  end_date = NULL,  
  aoi = NULL,  
  quality = TRUE,  
  max_granules = 200L  
)
```

Arguments

product: GEDI Earth Engine product name.

start_date, **end_date**: Optional dates for image collections.

aoi: Optional spatial filter.

quality: Apply available quality masks.

max_granules: Maximum number of vector granules to merge. Vector GEDI products are stored as folders of per-granule feature collections in Earth Engine, so use dates and/or an AOI to keep the request focused.

Value

An Earth Engine image, image collection, or feature collection.

gedifinder

Description

Find GEDI granules through NASA CMR

Usage

```
gedifinder(  
  product,  
  ul_lat,  
  ul_lon,  
  lr_lat,  
  lr_lon,  
  version = NULL,  
  daterange = NULL,  
  access = c("https", "s3", "opendap", "all"),  
  cloud_hosted = TRUE,  
  return = c("url", "table"),  
  page_size = 2000L,  
  cloud_computing = FALSE,  
  persist = TRUE  
)
```

Arguments

product: One of 'GEDI01_B', 'GEDI02_A', 'GEDI02_B', 'GEDI03', 'GEDI04_A', or 'GEDI04_B'.

ul_lat, ul_lon, lr_lat, lr_lon: Bounding coordinates in decimal degrees.

version: Product version. 'NULL' selects the current supported version.

daterange: Optional two-element date or date-time vector.

access: Requested link type: "https", "s3", "opendap", or "all".

cloud_hosted: Logical retained for backward compatibility. The current GEDI collection identifiers resolve to their Earthdata Cloud holdings.

return: Return a URL vector or a metadata table.

page_size: CMR page size.

cloud_computing: Logical. When 'TRUE', authenticate with Earthdata and return HTTPS granules marked for direct cloud streaming by 'openGEDI()' or the product-specific readLevel*() function.

persist: Logical; allow Earthaccess to persist an interactive login when 'cloud_computing = TRUE'.

Value

A character vector or data.table::data.table.

See Also

<<https://cmr.earthdata.nasa.gov/search/site/docs/search/api.html>>

gediWFMetrics

Description

Compute metrics from GEDI or simulated full waveforms

Usage

```
gediWFMetrics(  
  input,  
  outRoot = NULL,  
  rhRes = 5,  
  ground = TRUE,  
  rhoG = 0.4,  
  rhoC = 0.57,  
  ground_method = c("auto", "truth", "lowest_peak", "quantile"),  
  ...  
)
```

Arguments

input: A 'gedi.level1b' object or list of such objects.

outRoot: Optional output filename root.

rhRes: Relative-height percentile spacing.

ground: Logical; include the estimated ground elevation.

rhoG, rhoC: Ground and canopy reflectance.

ground_method: Ground estimator. "auto" uses the classified ground waveform when present and otherwise the lowest smoothed waveform peak.

...: Compatibility arguments from the original native implementation.

Value

A data.table::data.table with waveform metrics.

gediWFSimulator

Description

This portable implementation follows the core 'gediRat' algorithm: Gaussian footprint weighting, optional return-density normalization, vertical binning, pulse convolution, separation of LAS class 2 ground returns, and unit-integral normalization. Vectorized binning replaces the original point loop while avoiding its native GSL/GDAL build stack.

Usage

```
gediWFSimulator(  
  input,  
  output = tempfile(fileext = ".h5"),  
  ground = TRUE,  
  ascii = FALSE,  
  waveID = NULL,  
  coords = NULL,  
  listCoord = NULL,  
  gridBound = NULL,  
  gridStep = 30,  
  pSigma = -1,  
  pFWHM = 15,  
  fSigma = 5.5,  
  res = 0.15,  
  decimate = 1,  
  maxBins = 1024L,  
  countOnly = FALSE,  
  method = c("count", "intensity", "fraction"),  
  normalize = TRUE,  
  density_res = 1,  
  intensity_threshold = 6e-04,  
  noise = 0,  
  keepOld = FALSE,  
  seed = NULL,  
  ...  
)
```

Arguments

input: LAS/LAZ paths, a 'lidR::LAS', 'SpatVector', or table with X/Y/Z.

output: Output HDF5 path.

ground: Retained for compatibility; ground returns remain in the waveform.

ascii: Return/write a simple ASCII waveform instead of HDF5.

waveID: Optional waveform identifiers.

coords: A coordinate pair or matrix of footprint centers.

listCoord: Text file containing footprint centers.

gridBound: Optional 'c(xmin, xmax, ymin, ymax)' simulation grid.

gridStep: Grid spacing in input coordinate units.

pSigma: Pulse sigma in meters. A negative value derives it from 'pFWHM'.

pFWHM: Pulse full width at half maximum in nanoseconds.

fSigma: Footprint sigma in input coordinate units.

res: Vertical waveform resolution.

decimate: Probability of retaining an ALS return.

maxBins: Maximum number of waveform bins.

countOnly: Use count-based returns rather than ALS intensity. Gaussian footprint weights are still applied.

method: Waveform contribution: "count", "intensity", or "fraction". 'countOnly = TRUE' selects "count" for compatibility.

normalize: Correct for local ALS sampling density, as in the original simulator's default behavior.

density_res: Cell size used for sampling-density normalization.

intensity_threshold: Smallest retained Gaussian footprint weight.

noise: Relative Gaussian noise standard deviation. For example, 'noise = 0.03' adds noise with a standard deviation equal to three percent of the waveform maximum. Zero leaves the waveform unchanged.

keepOld: Do not overwrite an existing output file.

seed: Optional random seed.

...: Compatibility arguments accepted from the original simulator.

Value

A 'gedi.level1b' object, or a waveform table when 'ascii=TRUE'.

References

Hancock et al. (2019) \Sexpr[results=rd]{tools:::Rd_expr_doi("10.1029/2018EA000506")}

geomSampling

Description

GEDI polygon-stratified sampling specification

Usage

```
geomSampling(size, geom, split_id = NULL, lon = NULL, lat = NULL)
```

Arguments

size: Number or fraction per polygon.

geom: Polygon object.

split_id: Optional polygon ID column.

lon, lat: Optional coordinate column names.

get_catalog_id

Description

Return an Earth Engine catalog ID

Usage

```
get_catalog_id(x)
```

Arguments

x: A GEDI product name, catalog search row, or catalog ID.

Value

A character catalog ID.

getGEDITrack

Description

Standardizes footprint coordinates, acquisition order, and beam identifiers from any open point-level GEDI product or an extracted footprint table. The result can be plotted directly or passed to 'plot_gedi_orbit_animation()'.

Usage

```
getGEDITrack(  
  x,  
  lon = NULL,  
  lat = NULL,  
  time = NULL,  
  track = NULL,  
  beams = NULL,  
  every = 1L  
)
```

Arguments

x: A data frame, 'data.table', 'sf' or 'SpatVector' containing GEDI footprint coordinates, or an open 'gedi.level1b', 'gedi.level2a', 'gedi.level2b', or 'gedi.level4a' object.

lon, lat: Names of longitude and latitude columns. When 'NULL', common GEDI coordinate names are detected automatically.

time: Optional name of a column used to order footprints within each track. GEDI 'delta_time' and 'shot_number' columns are detected by default.

track: Optional name of a beam or track column. Separate tracks are colored independently.

beams: Optional beam names to read from an open GEDI object.

every: Keep every nth footprint after ordering. This is useful when plotting a complete orbit.

Value

A data.table::data.table with 'longitude', 'latitude', 'sequence', 'track', and available time and shot identifiers.

Examples

```
shots <- data.frame(  
  lon_lowestmode = seq(-44.2, -44.1, length.out = 10),  
  lat_lowestmode = seq(-13.8, -13.7, length.out = 10),  
  delta_time = seq_len(10), beam = "BEAM00000"  
)  
getGEDITrack(shots, every = 2)
```

getLevel1BGeo

Description

This function extracts Pulse Full Waveform Geolocations from GEDI 'gedi.level1b' data

Usage

```
getLevel1BGeo(  
  level1b,  
  select = c("elevation_bin0", "elevation_lastbin"),  
  beams = NULL  
)
```

Arguments

level1b: A 'gedi.level1b' object (output of 'getLevel1BGeo()' function).

select: A character vector specifying the fields to extract from GEDI Level1B data. If NULL, by default it will extract `_latitude_bin0_`, `_latitude_lastbin_`, `_longitude_bin0_`, `_longitude_lastbin_`, and `_shot_number_`. See details for more options.

beams: Optional beam names. 'NULL' reads every beam.

Details

Additional fields to be extracted from GEDI level 1B:

- * `_all_samples_sum_` Sum of all values within the 10 km range window.
- * `_beam_` Beam number Number.
- * `_channel_` Channel number.
- * `_master_frac_` Master time, fractional part.
- * `_master_int_` Master time, integer part.
- * `_noise_mean_corrected_` Noise mean.
- * `_noise_stddev_corrected_` Corrected noise standard deviation.
- * `_nsemean_even_` Noise mean of the beam's detector channel from even sub-converter.
- * `_nsemean_odd_` Noise mean of the beam's odd sub-converter.
- * `_rx_energy_` Integrated energy in receive (RX) waveform after subtracting the noise mean.
- * `_rx_offset_` Time interval from first stored sample to first downloaded RX sample.
- * `_rx_open_` Time interval from time 0 to first stored RX sample.
- * `_rx_sample_count_` The number of sample intervals (elements) in each RX waveform.
- * `_rx_sample_start_index_` The index in the rxwaveform dataset of the first element of each RX waveform starting at 1.
- * `_selection_stretchers_x_` Commanded number of samples added to the algorithm section on the left.
- * `_selection_stretchers_y_` Commanded number of samples added to the algorithm section on the right.
- * `_shot_number_` Unique shot identifier.
- * `_stale_return_flag_` Indicates that a "stale" cue point from the coarse search algorithm is being used.
- * `_th_left_used_` Count values for the left threshold used in fine search where two consecutive points at or above this value indicate pulse detection.
- * `_tx_egamplitude_` Amplitude of the extended Gaussian fit to the transmit (TX) waveform.
- * `_tx_egamplitude_error_` Error on `tx_egamplitude`.
- * `_tx_egbias_` Bias of the extended Gaussian fit to the TX waveform.
- * `_tx_egbias_error_` Error on `tx_egbias`.
- * `_tx_egflag_` Extended Gaussian fit status flag.
- * `_tx_eggamma_` Gamma value of the extended Gaussian fit to the TX waveform.

- * `_tx_eggamma_error_` Error on `tx_eggamma`.
- * `_tx_egsigma_` Sigma of the extended Gaussian fit to the TX waveform.
- * `_tx_egsigma_error_` Error on `tx_egsigma`.
- * `_tx_gloc_` Location (mean) of the Gaussian fit to the TX waveform.
- * `_tx_gloc_error_` Error on `tx_gloc`.
- * `_tx_pulseflag_` Set to 1 if a pulse is detected in the TX waveform.
- * `_tx_sample_count_` The number of sample intervals (elements) in each transmit waveform.
- * `_tx_sample_start_index_` The index in the `rxwaveform` dataset of the first element of each RX waveform starting at 1.
- * `_altitude_instrument_` Height of the instrument diffractive optical element (DOE) above the WGS84 ellipsoid.
- * `_altitude_instrument_error_` Error on `altitude_instrument`.
- * `_bounce_time_offset_bin0_` The difference between the TX time and the time at the start of the RX window.
- * `_bounce_time_offset_bin0_error_` Error on `bounce_time_offset_bin0`.
- * `_bounce_time_offset_lastbin_` The difference between the TX time and the time at the end of the RX window.
- * `_bounce_time_offset_lastbin_error_` Error on `bounce_time_offset_lastbin`.
- * `_degrade_` Greater than zero if the shot occurs during a degrade period, zero otherwise.
- * `_delta_time_` Transmit time of the shot, measured in seconds since 2018-01-01.
- * `_digital_elevation_model_` Digital elevation model height above the WGS84 ellipsoid.
- * `_elevation_bin0_` Height of the start of the RX window, relative to the WGS-84 ellipsoid.
- * `_elevation_bin0_error_` Error on `elevation_bin0`.
- * `_elevation_lastbin_` Height of the end of the RX window, relative to the WGS-84 ellipsoid.
- * `_elevation_lastbin_error_` Error on `elevation_lastbin`.
- * `_latitude_bin0_` Latitude of the start of the RX window.
- * `_latitude_bin0_error_` Error on `latitude_bin0`.
- * `_latitude_lastbin_` Latitude of the end of the RX window.
- * `_latitude_lastbin_error_` Error on `latitude_lastbin`.
- * `_latitude_instrument_` Latitude of the instrument diffractive optical element (DOE) at laser transmit time.
- * `_latitude_instrument_error_` Error on `latitude_instrument`.
- * `_local_beam_azimuth_` Azimuth of the unit pointing vector for the laser in the local East, North, Up (ENU) frame.
- * `_local_beam_azimuth_error_` Error on `local_beam_azimuth`.
- * `_local_beam_elevation_` Elevation of the unit pointing vector for the laser in the local ENU frame.
- * `_local_beam_elevation_error_` Error on `local_beam_elevation`.
- * `_longitude_bin0_` Longitude of the start of the RX window.
- * `_longitude_bin0_error_` Error on `longitude_bin0`.
- * `_longitude_lastbin_` Longitude of the end of the RX window.
- * `_longitude_lastbin_error_` Error on `longitude_lastbin`.
- * `_longitude_instrument_` Longitude of the instrument diffractive optical element (DOE) at laser transmit time.
- * `_longitude_instrument_error_` Error on `longitude_instrument`.
- * `_mean_sea_surface_` Mean sea surface height above the WGS84 ellipsoid, includes the geoid .
- * `_neutat_delay_derivative_bin0_` Change in neutral atmospheric delay per height change for the start of the RX window.
- * `_neutat_delay_derivative_lastbin_` Change in neutral atmospheric delay per height change for the end of the RX window.
- * `_neutat_delay_total_bin0_` Total neutral atmosphere delay correction (wet+dry) from the TX pulse to the start of the RX window.

- * `_neutat_delay_total_lastbin` Total neutral atmosphere delay correction (wet+dry) from the TX pulse to the end of the RX window.
- * `_range_bias_correction` The range bias applied to the range measurement.
- * `_shot_number` Unique shot identifier Number.
- * `_solar_azimuth` The azimuth of the sun position vector.
- * `_solar_elevation` The elevation of the sun position vector.
- * `_surface_type` Flags describing which surface types.
- * `_dynamic_atmosphere_correction` Dynamic Atmospheric Correction (DAC) includes inverted barometer (IB) effect.
- * `_geoid` Geoid height above WGS-84 reference ellipsoid.
- * `_tide_earth` Solid Earth tides.
- * `_tide_load` Load Tide - Local displacement due to Ocean Loading.
- * `_tide_ocean` Ocean Tides including diurnal and semi-diurnal, and longerperiod tides.
- * `_tide_ocean_pole` Oceanic surface rotational deformation due to polar motion.
- * `_tide_pole` Solid Earth Pole Tide. Rotational deformation due to polar motion.

Value

Returns an S4 object of class 'data.table::data.table' containing the GEDI Full Waveform Geolocations

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi01-b-002>>

Examples

```
# specify the path to GEDI level1B data (zip file)
outdir = tempdir()
level1B_fp_zip <- system.file("extdata",
                             "GEDI01_B_2019108080338_001964_T05337_02_003_01_sub.zip",
                             package="rGEDI")

# Unzipping GEDI level1B data
level1Bpath <- unzip(level1B_fp_zip, exdir = outdir)

# Reading GEDI level1B data (h5 file)
level1b <- readLevel1B(level1Bpath=level1Bpath)

# Extracting GEDI level1B geolocations
level1bGeo <- getLevel1BGeo(level1b, select=c("elevation_bin0", "elevation_lastbin"))
head(level1bGeo)

close(level1b)
```


getLevel1BWF

Description

This function extracts the full waveform of a given pulse from GEDI Level1B data.

Usage

```
getLevel1BWF(level1b, shot_number)
```

Arguments

level1b: A GEDI Level1B object (output of 'readLevel1B()' function). A S4 object of class "gedi.level1b".

shot_number: Shot number. A scalar representing the shot number of a giving pulse.

Details

Shot numbers can be extracted using readLevel1B function.

Value

Returns an S4 object of class "gedi.fullwaveform".

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi01-b-002>>

Examples

```
# Specifying the path to GEDI level1B data (zip file)
outdir = tempdir()
level1B_fp_zip <- system.file("extdata",
                              "GEDI01_B_2019108080338_001964_T05337_02_003_01_sub.zip",
                              package="rGEDI")

# Unzipping GEDI level1B data
level1Bpath <- unzip(level1B_fp_zip, exdir = outdir)

# Reading GEDI level1B data (h5 file)
level1b <- readLevel1B(level1Bpath=level1Bpath)

# Extracting GEDI full waveform for a giving shotnumber
wf <- getLevel1BWF(level1b, shot_number="19640521100108408")

# Plotting GEDI Full waveform
oldpar <- par()
par(mfrow = c(1,2), cex.axis = 1.5)
plot(wf, relative=FALSE, polygon=TRUE, type="l", lwd=2, col="forestgreen",
     xlab="Waveform Amplitude", ylab="Elevation (m)")

plot(wf, relative=TRUE, polygon=TRUE, type="l", lwd=2, col="forestgreen",
     xlab="Waveform Amplitude (%)", ylab="Elevation (m)")

par(oldpar)
close(level1b)
```

getLevel2AM

Description

This function extracts Elevation and Relative Height (RH) metrics from GEDI Level2A data.

Usage

```
getLevel2AM(level2a, beams = NULL, include_rh = TRUE)
```

Arguments

level2a: A GEDI Level2A object (output of 'readLevel2A()' function). An S4 object of class "gedi.level2a".

beams: Optional beam names. 'NULL' reads every beam.

include_rh: Logical. Include the 101 relative-height columns. Set to 'FALSE' for a faster metadata-only cloud read.

Details

Characteristics. Flag indicating likely invalid waveform (1=valid, 0=invalid).

- * _beam_ Beam identifier
- * _shot_number_ Shot number
- * _degrade_flag_ Flag indicating degraded state of pointing and/or positioning information
- * _quality_flag_ Flag simplifying selection of most useful data
- * _delta_time_ Transmit time of the shot since Jan 1 00:00 2018
- * _sensitivity_ Maximum canopy cover that can be penetrated
- * _solar_elevation_ Solar elevation
- * _lat_lowestmode_ Latitude of center of lowest mode
- * _lon_lowestmode_ Longitude of center of lowest mode
- * _elev_highestreturn_ Elevation of highest detected return relative to reference ellipsoid Meters
- * _elev_lowestmode_ Elevation of center of lowest mode relative to reference ellipsoid
- * _rh_ Relative height metrics at 1% interval

Value

Returns an S4 object of class data.table::data.table containing the elevation and relative heights metrics.

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi02-a-002>>

Examples

```
# Specifying the path to GEDI level2A data (zip file)
outdir = tempdir()
level2A_fp_zip <- system.file("extdata",
                              "GEDI02_A_2019108080338_001964_T05337_02_001_01_sub.zip",
                              package="rGEDI")

# Unzipping GEDI level2A data
level2Apath <- unzip(level2A_fp_zip, exdir = outdir)

# Reading GEDI level2A data (h5 file)
level2a <- readLevel2A(level2Apath=level2Apath)

# Extracting GEDI Elevation and Height Metrics
level2AM <- getLevel2AM(level2a)
head(level2AM)

close(level2a)
```

getLevel2BPAIProfile

Description

This function extracts the Plant Area Index (PAI) Profile from GEDI Level2B data.

Usage

```
getLevel2BPAIProfile(level2b, beams = NULL)
```

Arguments

level2b: A GEDI Level2B object (output of 'readLevel2B()' function). An S4 object of class "gedi.level2b".

beams: Optional beam names. 'NULL' reads every beam.

Details

Characteristics. Flag indicating likely invalid waveform (1=valid, 0=invalid).

- * `_beam_` Beam identifier
- * `_shot_number_` Shot number
- * `_algorithmrun_flag_` The L2B algorithm is run if this flag is set to 1 indicating data have sufficient waveform fidelity for L2B to run
- * `_l2b_quality_flag_` L2B quality flag
- * `_delta_time_` Transmit time of the shot since Jan 1 00:00 2018
- * `_lat_lowestmode_` Latitude of center of lowest mode
- * `_lon_lowestmode_` Longitude of center of lowest mode
- * `_elev_highestreturn_` Elevation of highest detected return relative to reference ellipsoid
- * `_elev_lowestmode_` Elevation of center of lowest mode relative to reference ellipsoid
- * `_height_lastbin_` Height of the last bin of the pgap_theta_z, relative to the ground
- * `_pai_z_` Plant Area Index profile

Value

Returns an S4 object of class `data.table::data.table` containing the elevation and relative heights.

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi02-b-002>>

Examples

```
# Specifying the path to GEDI level2B data (zip file)
outdir = tempdir()
level2B_fp_zip <- system.file("extdata",
                              "GEDI02_B_2019108080338_001964_T05337_02_001_01_sub.zip",
                              package="rGEDI")

# Unzipping GEDI level2A data
level2Bpath <- unzip(level2B_fp_zip, exdir = outdir)

# Reading GEDI level2B data (h5 file)
level2b <- readLevel2B(level2Bpath=level2Bpath)

# Extracting GEDI Plant Area Index (PAI) Profile (GEDI Level2B)
level2BPAIProfile <- getLevel2BPAIProfile(level2b)
head(level2BPAIProfile)

close(level2b)
```

getLevel2BPAVDProfile

Description

This function extracts the Plant Area Volume Density (PAVD) Profile from GEDI Level2B data.

Usage

```
getLevel2BPAVDProfile(level2b, beams = NULL)
```

Arguments

level2b: A GEDI Level2B object (output of 'readLevel2B()' function). An S4 object of class "gedi.level2b".

beams: Optional beam names. 'NULL' reads every beam.

Details

Characteristics. Flag indicating likely invalid waveform (1=valid, 0=invalid).

- * _beam_ Beam identifier
- * _shot_number_ Shot number
- * _algorithmrun_flag_ The L2B algorithm is run if this flag is set to 1 indicating data have sufficient waveform fidelity for L2B to run
- * _l2b_quality_flag_ L2B quality flag
- * _delta_time_ Transmit time of the shot since Jan 1 00:00 2018
- * _lat_lowestmode_ Latitude of center of lowest mode
- * _lon_lowestmode_ Longitude of center of lowest mode
- * _elev_highestreturn_ Elevation of highest detected return relative to reference ellipsoid
- * _elev_lowestmode_ Elevation of center of lowest mode relative to reference ellipsoid
- * _height_lastbin_ Height of the last bin of the pgap_theta_z, relative to the ground
- * _pavd_z_ Plant Area Volume Density profile

Value

Returns an S4 object of class data.table::data.table containing the Plant Area Volume Density Index.

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi02-b-002>>

Examples

```
# Specifying the path to GEDI level2B data (zip file)
outdir = tempdir()
level2B_fp_zip <- system.file("extdata",
                              "GEDI02_B_2019108080338_001964_T05337_02_001_01_sub.zip",
                              package="rGEDI")

# Unzipping GEDI level2A data
level2Bpath <- unzip(level2B_fp_zip, exdir = outdir)

# Reading GEDI level2B data (h5 file)
level2b <- readLevel2B(level2Bpath=level2Bpath)

# Extracting GEDI Plant Area Volume Density (PAVD) Index
level2BPAVDProfile <- getLevel2BPAVDProfile(level2b)
head(level2BPAVDProfile)

close(level2b)
```

getLevel2BVPM

Description

This function extracts information from GEDI Level2B data: Total Plant Area Index, Foliage Height Diversity, Foliage Clumping Index, Total Gap Probability (theta), and Total canopy cover.

Usage

```
getLevel2BVPM(  
  level2b,  
  cols = c("beam", "shot_number", "algorithmrun_flag", "l2b_quality_flag", "delta_time",  
    "sensitivity", "solar_elevation", "latitude_lastbin", "latitude_bin0",  
    "longitude_bin0", "longitude_lastbin", "elev_highestreturn", "elev_lowestmode",  
    "rh100", "pai", "fhd_normal", "omega", "pgap_theta", "cover"),  
  beams = NULL  
)
```

Arguments

level2b: A GEDI Level2B object (output of 'readLevel2B()' function). An S4 object of class "gedi.level2b".

cols: A character vector containing the list of columns to be extracted. See the default columns in the description.

beams: Optional beam names. 'NULL' reads every beam.

Details

These are the biophysical variables and additional information extracted by default:

- * _beam_ Beam identifier
- * _shot_number_ Shot number
- * _algorithmrun_flag_ The L2B algorithm is run if this flag is set to 1 indicating data have sufficient waveform fidelity for L2B to run
- * _l2b_quality_flag_ L2B quality flag
- * _delta_time_ Transmit time of the shot since Jan 1 00:00 2018
- * _sensitivity_ Maximum canopy cover that can be penetrated
- * _solar_elevation_ Solar elevation
- * _latitude_lastbin_ Latitude of last bin of the pgap_theta_z, interpolated from L1B waveform coordinate
- * _latitude_bin0_ Latitude of first bin of the pgap_theta_z, interpolated from L1B waveform coordinate
- * _elev_highestreturn_ Elevation of highest detected return relative to reference ellipsoid
- * _elev_lowestmode_ Elevation of center of lowest mode relative to reference ellipsoid
- * _rh100_ RH100 slice
- * _pai_ Total Plant Area Index
- * _fhd_normal_ Foliage Height Diversity
- * _omega_ Foliage Clumping Index
- * _pgap_theta_ Total Gap Probability (theta)
- * _cover_ Total canopy cover

Every other columns in the GEDI2B product are also available, you can specify each column by using the 'cols' parameter.

Value

Returns an S4 object of class data.table::data.table containing the Vegetation Profile Biophysical Variables.

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi02-b-002>>

Examples

```
# Specifying the path to GEDI level2B data (zip file)  
outdir = tempdir()
```

```

level2B_fp_zip <- system.file("extdata",
                              "GEDI02_B_2019108080338_001964_T05337_02_001_01_sub.zip",
                              package="rGEDI")

# Unzipping GEDI level2A data
level2Bpath <- unzip(level2B_fp_zip, exdir = outdir)

# Reading GEDI level2B data (h5 file)
level2b<-readLevel2B(level2Bpath=level2Bpath)

# Extracting GEDI Vegetation Profile Biophysical Variables
level2BVPM<-getLevel2BVPM(level2b, cols=c("beam", "shot_number"))
head(level2BVPM)

close(level2b)

```

getLevel4A

Description

Extract GEDI Level 4A footprint metrics

Usage

```
getLevel4A(  
  level4a,  
  cols = .level4a_default_columns,  
  quality = FALSE,  
  beams = NULL  
)
```

Arguments

level4a: A 'gedi.level4a' object returned by 'readLevel4A()'.

cols: Character vector of fields to return. 'NULL' returns all datasets shared by the selected beams.

quality: Logical. If 'TRUE', retain observations accepted by the current Release 3 quality fields ('l4a_quality_flag_rel3', 'degrade_include_flag', and 'elev_highestreturn_outlier_flag') when present. Legacy Release 2 quality fields remain supported.

beams: Optional character vector of GEDI beam names.

Value

A data.table::data.table with one row per footprint.

getTileUrl

Description

Get an Earth Engine image tile URL

Usage

```
getTileUrl(image, vis = list())
```

Arguments

image: Earth Engine image.

vis: Visualization parameter list.

Value

Character tile URL.

glcmTexture

Description

Earth Engine gray-level co-occurrence textures

Usage

```
glcmTexture(x, size = 1L, kernel = NULL, average = TRUE)
```

Arguments

x: Earth Engine integer image.

size: Neighborhood size.

kernel: Optional Earth Engine kernel.

average: Average directional metrics.

Value

An Earth Engine image.

gridSampling

Description

GEDI grid-stratified sampling specification

Usage

```
gridSampling(size, grid_size, lon = NULL, lat = NULL)
```

Arguments

size: Number or fraction per cell.

grid_size: Grid size in decimal degrees.

lon, lat: Optional coordinate column names.

gridStatsLevel2AM

Description

This function computes a series of user defined descriptive statistics within each grid cell for GEDI derived Elevation and Height Metrics (Level2A)

Usage

```
gridStatsLevel2AM(level2AM, func, res)
```

Arguments

level2AM: A GEDI Level2AM object (output of 'getLevel2AM()' function). An S4 object of class "data.table".

func: The function(s) to be applied to each cell

res: Spatial resolution in decimal degrees for the output stars raster layer

Value

Return a stars raster layer(s) of selected GEDI Elevation and Height Metric(s)

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi02-a-002>>

Examples

```
## Not run:

# specify the path to GEDI level2A data (zip file)
outdir <- tempdir()
level2A_fp_zip <- system.file("extdata",
  "GEDI02_A_2019108080338_001964_T05337_02_001_01_sub.zip",
  package = "rGEDI"
)

# Unzipping GEDI level2A data
level2Apath <- unzip(level2A_fp_zip, exdir = outdir)

# Reading GEDI level2A data (h5 file)
level2a <- readLevel2A(level2Apath = level2Apath)

# Get GEDI derived Elevation and Height Metrics
level2AM <- getLevel2AM(level2a)
head(level2AM)

#' Define your own function
mySetOfMetrics <- function(x) {
  metrics <- list(
    min = min(x), # Min of z
    max = max(x), # Max of z
    mean = mean(x), # Mean of z
    sd = sd(x) # Sd of z
  )
  return(metrics)
}

#' Computing a serie of GEDI metrics
ZTstats <- gridStatsLevel2AM(
  level2AM = level2AM,
  func = mySetOfMetrics(elev_highestreturn),
  res = 0.005
)
if (interactive()) plot(ZTstats)

if (interactive()) {
  # Computing the maximum of RH100 only
  maxRH100 <- gridStatsLevel2AM(level2AM, mySetOfMetrics(rh100), res = 0.0005)
  plot(maxRH100)

  # Computing the mean of ZG only
  ZGmean <- gridStatsLevel2AM(level2AM, mean(elev_lowestmode), res = 0.005)
  plot(ZGmean)
}
close(level2a)
```

```
## End(Not run)
```

gridStatsLevel2BVPM

Description

This function computes a series of user defined descriptive statistics within each grid cell for GEDI derived Canopy Cover and Vertical Profile Metrics (Level2B)

Usage

```
gridStatsLevel2BVPM(level2BVPM, func, res)
```

Arguments

level2BVPM: A GEDI Level2AM object (output of 'getLevel2BVPM()' function). An S4 object of class "data.table".

func: The function(s) to be applied to each cell

res: Spatial resolution in decimal degrees for the output stars raster layer

Value

Returns a stars raster layer(s) of selected GEDI Canopy Cover and Vertical Profile Metric(s)

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi02-b-002>>

Examples

```
## Not run:

# specify the path to GEDI level2B data (zip file)
outdir <- tempdir()
level2B_fp_zip <- system.file("extdata",
  "GEDI02_B_2019108080338_001964_T05337_02_001_01_sub.zip",
  package = "rGEDI"
)

# Unzipping GEDI level2A data
level2Bpath <- unzip(level2B_fp_zip, exdir = outdir)

# Reading GEDI level2B data (h5 file)
level2b <- readLevel2B(level2Bpath = level2Bpath)

# Get GEDI derived Canopy Cover and Vertical Profile Metrics
level2BVPM <- getLevel2BVPM(level2b)
head(level2BVPM)

#' Define your own function
mySetOfMetrics <- function(x) {
  metrics <- list(
    min = min(x), # Min of z
    max = max(x), # Max of z
    mean = mean(x), # Mean of z
    sd = sd(x) # Sd of z
  )
  return(metrics)
}

#' Computing a serie of statistics of GEDI derived canopy cover
cover_stats <- gridStatsLevel2BVPM(
  level2BVPM = level2BVPM,
  func = mySetOfMetrics(cover),
  res = 0.005
)
plot(cover_stats)

#' Computing the max of the Total Plant Area Index only
pai_max <- gridStatsLevel2BVPM(level2BVPM = level2BVPM, func = max(pai), res = 0.005)
plot(pai_max)

#' Computing the Foliage Height Diversity Index only
fhd_mean <- gridStatsLevel2BVPM(level2BVPM = level2BVPM, func = mean(fhd_normal), res = 0.005)
plot(fhd_mean)

close(level2b)
## End(Not run)
```

gridStatsLevel4A

Description

Grid GEDI Level 4A footprint metrics

Usage

```
gridStatsLevel4A(level4A, metric = "agbd", fun = mean, res = 0.01, ...)
```

Arguments

level4A: A table returned by 'getLevel4A()'.

metric: Metric column name.

fun: Aggregation function.

res: Output resolution in decimal degrees.

...: Additional arguments passed to 'fun'.

Value

A 'terra::SpatRaster'.

map_create

Description

Apply an Earth Engine model to a predictor image

Usage

```
map_create(model, stack, aoi = NULL, name = "prediction")
```

Arguments

model: A trained Earth Engine classifier or regressor.

stack: Earth Engine predictor image.

aoi: Optional area to clip.

name: Output band name.

Value

An Earth Engine image.

map_download

Description

Download an Earth Engine image

Usage

```
map_download(  
  image,  
  filename,  
  region,  
  scale = 30,  
  crs = "EPSG:4326",  
  overwrite = FALSE  
)
```

Arguments

image: Earth Engine image.

filename: Destination GeoTIFF path.

region: Area of interest.

scale: Pixel size in meters.

crs: Output CRS.

overwrite: Logical.

Value

Invisibly returns 'filename'.

map_view

Description

Display Earth Engine layers in Leaflet

Usage

```
map_view(layers, vis = list(), aoi = NULL)
```

Arguments

layers: Named list of Earth Engine images.

vis: Named list of visualization parameter lists.

aoi: Optional spatial object used to fit map bounds.

Value

A Leaflet widget.

mosaicLevel3

Description

Mosaic GEDI Level 3 rasters

Usage

```
mosaicLevel3(x, fun = "mean", filename = "", overwrite = FALSE)
```

Arguments

x: Raster paths, a 'SpatRaster', or a list of rasters.

fun: Function used for overlapping cells.

filename: Optional output filename.

overwrite: Logical.

Value

A 'terra::SpatRaster'.

mosaicLevel4B

Description

Mosaic GEDI Level 4B rasters

Usage

```
mosaicLevel4B(x, fun = "mean", filename = "", overwrite = FALSE)
```

Arguments

x: Raster paths, a 'SpatRaster', or a list of rasters.

fun: Function used for overlapping cells.

filename: Optional output filename.

overwrite: Logical.

Value

A 'terra::SpatRaster'.

openGEDI

Description

Raster products are opened with GDAL through 'terra::rast()'. HDF5 cloud granules are opened lazily with Python 'earthaccess' and 'h5py'.

Usage

```
openGEDI(x, product = NULL, stream = TRUE, cache_dir = tempdir())
```

Arguments

x: Local path, HTTPS URL, S3 URI, OPeNDAP URL, or an Earthaccess granule.

product: Optional product identifier.

stream: Logical; stream raster data rather than download it.

cache_dir: Download directory for non-streamed data.

Value

A 'SpatRaster', 'gedi.cloud_h5', or local GEDI object.

plot

Description

For 'gedi.fullwaveform': will plot the full waveform

Usage

```
## S4 method for signature 'gedi.fullwaveform,missing'  
plot(x, relative = FALSE, polygon = FALSE, ...)
```

Arguments

x: An object of class 'gedi.fullwaveform' (output of 'getLevel1BWF()' function)

relative: if TRUE, the Waveform Amplitude will be showed in percentage (%)

polygon: if TRUE, the polygon will be added to the plot

...: will be passed to the main plot

Value

No return value

Examples

```
# Specifying the path to GEDI level1B data (zip file)  
outdir <- tempdir()  
level1B_fp_zip <- system.file("extdata",  
  "GEDI01_B_2019108080338_001964_T05337_02_003_01_sub.zip",  
  package = "rGEDI"  
)  
  
# Unzipping GEDI level1B data  
level1Bpath <- unzip(level1B_fp_zip, exdir = outdir)  
  
# Reading GEDI level1B data (h5 file)  
level1b <- readLevel1B(level1Bpath = level1Bpath)  
  
# Extracting GEDI Full-Waveform  
wf <- getLevel1BWF(level1b, shot_number = "19640521100108408")  
  
# Plotting GEDI Full-waveform  
oldpar <- par()  
par(mfrow = c(1, 2), cex.axis = 1.5)  
plot(wf,  
  relative = FALSE, polygon = TRUE, type = "l", lwd = 2, col = "forestgreen",  
  xlab = "", ylab = "Elevation (m)"  
)  
  
rGEDI::plot(wf,  
  relative = TRUE, polygon = TRUE, type = "l", lwd = 2, col = "forestgreen",  
  xlab = "Waveform Amplitude (%)", ylab = "Elevation (m)"  
)  
  
par(oldpar)  
close(level1b)
```

plot.gedi_var_selection

Description

Plot GEDI variable selection results

Usage

```
## S3 method for class 'gedi_var_selection'  
plot(x, which = c("importance", "rfe"), ...)
```

Arguments

x: A 'gedi_var_selection' object.

which: Plot variable importance or RFE error.

...: Additional graphical parameters.

Value

The object, invisibly.

plot_gedi_orbit_animation

Description

Creates a self-contained HTML animation or an animated GIF from GEDI footprint coordinates. HTML output uses an interactive orthographic globe; GIF output shows progressive reference-ground-track playback.

Usage

```
plot_gedi_orbit_animation(  
  x,  
  lon = NULL,  
  lat = NULL,  
  time = NULL,  
  track = NULL,  
  output_file = tempfile("rGEDI-orbit-", fileext = ".html"),  
  title = "GEDI orbit animation",  
  duration = 18,  
  launch = interactive()  
)
```

Arguments

x: A data frame, 'data.table', 'sf' or 'SpatVector' containing GEDI footprint coordinates, or an open 'gedi.level1b', 'gedi.level2a', 'gedi.level2b', or 'gedi.level4a' object.

lon, lat: Names of longitude and latitude columns. When 'NULL', common GEDI coordinate names are detected automatically.

time: Optional name of a column used to order footprints within each track. GEDI 'delta_time' and 'shot_number' columns are detected by default.

track: Optional name of a beam or track column. Separate tracks are colored independently.

output_file: Path of the '.html' or '.gif' file to create. GIF output requires the suggested 'gifski' package.

title: Title shown above the animation.

duration: Duration of one animation cycle in seconds.

launch: Open the animation in the default browser after it is written.

Value

The normalized path to the HTML file, invisibly.

Examples

```
orbit <- data.frame(  
  longitude = seq(-60, -45, length.out = 30),  
  latitude = seq(-8, 5, length.out = 30),  
  delta_time = seq_len(30), beam = "BEAM0101"  
)  
html <- plot_gedi_orbit_animation(  
  orbit, output_file = tempfile(fileext = ".html")  
)  
file.exists(html)
```

plotLevel3

Description

Plot GEDI Level 3 metrics

Usage

```
plotLevel3(x, ...)
```

Arguments

x: A raster path or 'SpatRaster'.

...: Passed to 'terra::plot()'.

Value

Invisibly returns the raster.

plotLevel4A

Description

Plot GEDI Level 4A footprint biomass

Usage

```
plotLevel4A(level4A, metric = "agbd", ...)
```

Arguments

level4A: A table returned by 'getLevel4A()'.

metric: Metric column to display.

...: Additional arguments passed to 'graphics::plot()'.

Value

Invisibly returns 'level4A'.

plotLevel4B

Description

Plot GEDI Level 4B metrics

Usage

```
plotLevel4B(x, ...)
```

Arguments

x: A raster path or 'SpatRaster'.

...: Passed to 'terra::plot()'.

Value

Invisibly returns the raster.

plotPAIProfile

Description

This functions plots Plant Area Index (PAI) Profile (GEDI level2B)

Usage

```
plotPAIProfile(level2BPAIProfile, beam = "BEAM0101", elev = TRUE)
```

Arguments

level2BPAIProfile: A GEDI Level2B object (output of 'getLevel2BPAIProfile()' function). An S4 object of class "data.table".

beam: Select GEDI beam. Default is "BEAM0101". See details section.

elev: If TRUE, elevation will be used for plotting the PAI profile. Otherwise, height will be used instead.

Details

list of GEDI beams. See the output of 'getLevel2BPAIProfile()' function.

```
* _BEAM0000_  
* _BEAM0001_  
* _BEAM0010_  
* _BEAM0011_  
* _BEAM0101_  
* _BEAM0110_  
* _BEAM1000_  
* _BEAM1011_
```

Value

Returns a ggplot object. See ggplot2::ggplot package.

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi02-b-002>>

Examples

```
# specify the path to GEDI level2B data (zip file)  
outdir = tempdir()  
level2B_fp_zip <- system.file("extdata",  
                              "GEDI02_B_2019108080338_001964_T05337_02_001_01_sub.zip",  
                              package="rGEDI")  
  
# Unzipping GEDI level2A data  
level2Bpath <- unzip(level2B_fp_zip, exdir = outdir)  
  
# Reading GEDI level2B data (h5 file)  
level2b <- readLevel2B(level2Bpath = level2Bpath)  
  
# Get Plant Area Volume Density profile  
level2BPAIProfile <- getLevel2BPAIProfile(level2b)  
  
# Plot Level2B PAI Profile  
gprofile <- plotPAIProfile(level2BPAIProfile, beam = "BEAM0101", elev = TRUE)  
  
close(level2b)
```

plotPAVDProfile

Description

This functions plots Plant Area Volume Density profile (GEDI level2B)

Usage

```
plotPAVDProfile(level2BPAVDProfile, beam = "BEAM0101", elev = TRUE)
```

Arguments

level2BPAVDProfile: A GEDI Level2B object (output of 'getLevel2BPAVDProfile()' function). An S4 object of class "data.table".

beam: Select GEDI beam. Default is "BEAM0101". See details section.

elev: If TRUE, elevation will be used for plotting the PAVD profile. Otherwise, height will be used instead.

Details

list of GEDI beams. See the output of 'getLevel2BPAVDProfile()' function.

```
* _BEAM0000_  
* _BEAM0001_  
* _BEAM0010_  
* _BEAM0011_  
* _BEAM0101_  
* _BEAM0110_  
* _BEAM1000_  
* _BEAM1011_
```

Value

Returns a ggplot object. See ggplot2::ggplot package.

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi02-b-002>>

Examples

```
# specify the path to GEDI level2B data (zip file)  
outdir = tempdir()  
level2B_fp_zip <- system.file("extdata",  
                              "GEDI02_B_2019108080338_001964_T05337_02_001_01_sub.zip",  
                              package="rGEDI")  
  
# Unzipping GEDI level2A data  
level2Bpath <- unzip(level2B_fp_zip, exdir = outdir)  
  
# Reading GEDI level2B data (h5 file)  
level2b <- readLevel2B(level2Bpath = level2Bpath)  
  
# Get Plant Area Volume Density profile  
level2BPAVDProfile <- getLevel2BPAVDProfile(level2b)  
  
# Plot Level2B PAVD Profile  
gprofile <- plotPAVDProfile(level2BPAVDProfile, beam = "BEAM0101", elev = TRUE)  
  
close(level2b)
```

plotWFMetrics

Description

Plots the waveform with overlaid RH metrics

Usage

```
plotWFMetrics(level1b, level2a, shot_number, rh=c(25, 50, 75),...)
```

Arguments

level1b: A GEDI Level1B object (output of 'readLevel1B()' function). An S4 object of class "gedi.level1b".

level2a: A GEDI Level2A object (output of 'readLevel2A()' function). An S4 object of class "gedi.level2a".

shot_number: Shot number. A scalar representing the shot number of a giving pulse.

rh: Integer vector. Specify which RH metrics to plot except rh0 and rh100, default c(25, 50, 75).

...: Will be passed to the main plot.

Value

Nothing

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi02-b-002>>

Examples

```
# specify the path to GEDI level1B and Level2A data (zip file)
outdir <- tempdir()
level1B_fp_zip <- system.file("extdata",
  "GEDI01_B_2019108080338_001964_T05337_02_003_01_sub.zip",
  package = "rGEDI"
)

level2A_fp_zip <- system.file("extdata",
  "GEDI02_A_2019108080338_001964_T05337_02_001_01_sub.zip",
  package = "rGEDI"
)

# Unzipping GEDI level1B data
level1Bpath <- unzip(level1B_fp_zip, exdir = outdir)
level2Apath <- unzip(level2A_fp_zip, exdir = outdir)

# Reading GEDI level1B and Level2A data (h5 file)
level1b <- readLevel1B(level1Bpath = level1Bpath)
level2a <- readLevel2A(level2Apath = level2Apath)

shot_number <- "19640521100108408"

plotWFMetrics(level1b, level2a, shot_number, rh = c(25, 50, 75, 90))

close(level1b)
close(level2a)
```

polyStatsLevel2AM

Description

Computes a Series of Statistics from GEDI derived Elevation and Height Metrics (Level2A) within a given area defined or not by a polygon

Usage

```
polyStatsLevel2AM(level2AM, func, id=NULL)
```

Arguments

level2AM: A GEDI Level2AM object (output of 'getLevel2AM()' function). An S4 object of class "data.table".

func: The function to be applied for computing the defined statistics

id: A vector containing the polygon id for each GEDI observation. Default is NULL

Value

Returns an S4 object of class data.table::data.table Containing Statistics of GEDI level2A defined metrics

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi02-a-002>>

Examples

```
# Specifying the path to GEDI level2A data (zip file)
outdir <- tempdir()
level2A_fp_zip <- system.file("extdata",
  "GEDI02_A_2019108080338_001964_T05337_02_001_01_sub.zip",
  package = "rGEDI"
)

# Unzipping GEDI level2A data
level2Apath <- unzip(level2A_fp_zip, exdir = outdir)

# Reading GEDI level2A data (h5 file)
level2a <- readLevel2A(level2Apath = level2Apath)

# Specifying the path to shapefile
polygon_filepath <- system.file("extdata", "stands_cerrado.shp", package = "rGEDI")

# Reading shapefile as sf object
library(sf)
polygon <- sf::st_read(polygon_filepath)

# Extracting GEDI Elevation and Relative Metrics (level2A)
level2AM <- getLevel2AM(level2a)
head(level2AM)

# Clipping GEDI data by geometry
level2AM_clip <- clipLevel2AMGeometry(level2AM, polygon, split_by = "id")

#' Define your own function
mySetOfMetrics <- function(x) {
  metrics <- list(
    min = min(x), # Min of x
    max = max(x), # Max of x
    mean = mean(x), # Mean of x
    sd = sd(x) # Sd of x
  )
  return(metrics)
}

# Computing the maximum of RH100
RH100max <- polyStatsLevel2AM(level2AM_clip, func = max(rh100))

# Computing the maximum of RH100 stratified by polygon
RH100max_poly <- polyStatsLevel2AM(level2AM_clip, func = max(rh100), id = "id")

# Computing a serie statistics for GEDI metrics stratified by polygon
RH100metrics <- polyStatsLevel2AM(level2AM_clip,
  func = mySetOfMetrics(rh100),
  id = "id"
)
head(RH100metrics)
```

`close(level2a)`

polyStatsLevel2BVPM

Description

Computes a Series of Statistics of GEDI derived Canopy Cover and Vertical Profile metrics within a given area defined or not by a polygon

Usage

```
polyStatsLevel2BVPM(level2BVPM, func, id=NULL)
```

Arguments

level2BVPM: A GEDI Level2BVPM object (output of 'getLevel2BVPM()' function). An S4 object of class "data.table".

func: The function to be applied for computing the defined statistics

id: A vector containing the polygon id for each GEDI observation. Default is NULL

Value

Returns an S4 object of class data.table::data.table Containing Statistics of GEDI level2BVPM defined metrics

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi02-b-002>>

Examples

```
# Specifying the path to GEDI level2B data (zip file)
outdir <- tempdir()
level2B_fp_zip <- system.file("extdata",
  "GEDI02_B_2019108080338_001964_T05337_02_001_01_sub.zip",
  package = "rGEDI"
)

# Unzipping GEDI level2A data
level2Bpath <- unzip(level2B_fp_zip, exdir = outdir)

# Reading GEDI level2B data (h5 file)
level2b <- readLevel2B(level2Bpath = level2Bpath)

# Specifying the path to shapefile
polygon_filepath <- system.file("extdata", "stands_cerrado.shp", package = "rGEDI")

# Reading shapefile as sf object
library(sf)
polygon <- sf::st_read(polygon_filepath)

# Extracting GEDI Canopy Cover and Vertical Profile Metrics
level2BVPM <- getLevel2BVPM(level2b)
head(level2BVPM)

# Clipping GEDI data by geometry
level2BVPM_clip <- clipLevel2BVPMGeometry(level2BVPM, polygon, split_by = "id")

# Define your own function
mySetOfMetrics <- function(x) {
  metrics <- list(
    min = min(x), # Min of x
    max = max(x), # Max of x
    mean = mean(x), # Mean of x
    sd = sd(x) # Sd of x
  )
  return(metrics)
}

# Computing the max of the Total Plant Area Index
pai_max <- polyStatsLevel2BVPM(level2BVPM_clip, func = max(pai), id = NULL)
pai_max

# Computing the max of the Total Plant Area Index stratified by polygon
pai_max_poly <- polyStatsLevel2BVPM(level2BVPM_clip, func = max(pai), id = "poly_id")
head(pai_max_poly)

# Computing the serie of statistics of canopy cover stratified by polygon
cover_metrics <- polyStatsLevel2BVPM(level2BVPM_clip,
  func = mySetOfMetrics(cover),
  id = "id"
```



```
)  
head(cover_metrics)  
close(level2b)
```

polyStatsLevel3

Description

Polygon summaries of GEDI Level 3 metrics

Usage

```
polyStatsLevel3(x, y, fun = mean, na.rm = TRUE, ...)
```

Arguments

x: A raster path or 'SpatRaster'.

y: Points or polygons accepted by 'terra::vect()'.

fun: Summary function.

na.rm: Logical.

...: Passed to 'terra::extract()'.

Value

A data frame.

polyStatsLevel4A

Description

Polygon statistics for GEDI Level 4A footprints

Usage

```
polyStatsLevel4A(level4A, polygon, metric = "agbd", fun = mean, id = NULL, ...)
```

Arguments

level4A: A table returned by 'getLevel4A()'.

polygon: An 'sf', 'sfc', or 'SpatVector' polygon object.

metric: Metric column name.

fun: Aggregation function.

id: Optional polygon ID field.

...: Additional arguments passed to 'fun'.

Value

A data table with one row per polygon.

polyStatsLevel4B

Description

Polygon summaries of GEDI Level 4B metrics

Usage

```
polyStatsLevel4B(x, y, fun = mean, na.rm = TRUE, ...)
```

Arguments

x: A raster path or 'SpatRaster'.

y: Points or polygons accepted by 'terra::vect()'.

fun: Summary function.

na.rm: Logical.

...: Passed to 'terra::extract()'.

Value

A data frame.

predictGEDI

Description

Predict GEDI footprint values

Usage

```
predictGEDI(model, data, name = "prediction")
```

Arguments

model: A 'gedi_model' or fitted R model.

data: Predictor table.

name: Output column name.

Value

A data table containing input data and predictions.

predictGEDIH5

Description

Predict values and write them to HDF5

Usage

```
predictGEDIH5(model, data, output, dataset = "prediction")
```

Arguments

model: A 'gedi_model' or fitted R model.

data: Predictor table.

output: Output HDF5 filename.

dataset: Dataset name.

Value

Invisibly returns 'output'.

projectLevel3

Description

Project a GEDI Level 3 raster

Usage

```
projectLevel3(x, crs, method = "bilinear", ...)
```

Arguments

x: A raster path or 'SpatRaster'.

crs: Target CRS.

method: Resampling method.

...: Passed to 'terra::project()'.

Value

A 'terra::SpatRaster'.

projectLevel4B

Description

Project a GEDI Level 4B raster

Usage

```
projectLevel4B(x, crs, method = "bilinear", ...)
```

Arguments

x: A raster path or 'SpatRaster'.

crs: Target CRS.

method: Resampling method.

...: Passed to 'terra::project()'.

Value

A 'terra::SpatRaster'.

randomSampling

Description

GEDi random sampling specification

Usage

```
randomSampling(size)
```

Arguments

size: Number or fraction of observations to select.

rasterizeGEDI

Description

Rasterize GEDI footprint data

Usage

```
rasterizeGEDI(x, metric, res = 0.01, fun = mean, lon = NULL, lat = NULL, ...)
```

Arguments

x: GEDI footprint table.

metric: Metric column.

res: Resolution in decimal degrees.

fun: Aggregation function.

lon, lat: Optional coordinate columns.

...: Passed to 'terra::rasterize()'.

Value

A 'terra::SpatRaster'.

rasterizeLevel4A

Description

Rasterize GEDI Level 4A footprints

Usage

```
rasterizeLevel4A(  
  level4A,  
  metric = "agbd",  
  fun = mean,  
  res = 0.01,  
  filename = "",  
  overwrite = FALSE,  
  ...  
)
```

Arguments

level4A: A table returned by 'getLevel4A()'.

metric: Metric column name.

fun: Aggregation function.

res: Output resolution in decimal degrees.

filename: Optional output GeoTIFF path.

overwrite: Logical; overwrite 'filename' when it exists.

...: Additional arguments passed to 'fun'.

Value

A 'terra::SpatRaster'.

rasterSampling

Description

GEDI raster-stratified sampling specification

Usage

```
rasterSampling(size, raster, lon = NULL, lat = NULL)
```

Arguments

size: Number or fraction per raster class.

raster: A categorical 'SpatRaster'.

lon, lat: Optional coordinate column names.

readGEDICloud

Description

Open a GEDI HDF5 granule through Earthaccess

Usage

```
readGEDICloud(x, product = NULL, cache_dir = tempdir())
```

Arguments

x: URL, S3 URI, or Earthaccess granule result.

product: Optional product identifier.

cache_dir: Used when cloud modules are unavailable and a URL is downloaded.

Value

An object of class 'gedi.cloud_h5'.

readLevel1B

Description

This function reads GEDI level1B products: geolocated Waveforms

Usage

```
readLevel1B(level1Bpath)
```

Arguments

level1Bpath: Local file path or Earthdata Cloud URL pointing to a GEDI Level 1B HDF5 granule.

Value

Returns an S4 object of class 'gedi.level1b' containing GEDI level1B data.

See Also

'hdf5r::H5File' in the `_hdf5r_` package and <<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi01-b-002>>

Examples

```
# Specifying the path to GEDI level1B data (zip file)
outdir = tempdir()
level1B_fp_zip <- system.file("extdata",
                              "GEDI01_B_2019108080338_001964_T05337_02_003_01_sub.zip",
                              package="rGEDI")

# Unzipping GEDI level1B data
level1Bpath <- unzip(level1B_fp_zip, exdir = outdir)

# Reading GEDI level1B data (h5 file)
level1b <- readLevel1B(level1Bpath=level1Bpath)

close(level1b)
```

readLevel2A

Description

This function reads GEDI level2A products: ground elevation, canopy top height, and relative heights (RH).

Usage

```
readLevel2A(level2Apath)
```

Arguments

level2Apath: Local file path or Earthdata Cloud URL pointing to a GEDI Level 2A HDF5 granule.

Value

Returns an S4 object of class 'gedi.level2a' containing GEDI level2A data.

See Also

<<https://www.earthdata.nasa.gov/data/catalog/lpcloud-gedi02-a-002>>

Examples

```
# Specifying the path to GEDI level2A data (zip file)
outdir = tempdir()
level2A_fp_zip <- system.file("extdata",
                             "GEDI02_A_2019108080338_001964_T05337_02_001_01_sub.zip",
                             package="rGEDI")

# Unzipping GEDI level2A data
level2Apath <- unzip(level2A_fp_zip, exdir = outdir)

# Reading GEDI level2A data (h5 file)
level2a <- readLevel2A(level2Apath=level2Apath)

close(level2a)
```

readLevel2B

Description

This function reads GEDI level2B products: canopy cover, Plant Area Index (PAI), Plant Area Volume Density (PAVD), and Foliage Height Diversity (FHD).

Usage

```
readLevel2B(level2Bpath)
```

Arguments

level2Bpath: Local file path or Earthdata Cloud URL pointing to a GEDI Level 2B HDF5 granule.

Value

Returns an S4 object of class 'gedi.level2b' containing GEDI level2B data.

Examples

```
# Specifying the path to GEDI level2B data (zip file)
outdir = tempdir()
level2B_fp_zip <- system.file("extdata",
                              "GEDI02_B_2019108080338_001964_T05337_02_001_01_sub.zip",
                              package="rGEDI")

# Unzipping GEDI level2A data
level2Bpath <- unzip(level2B_fp_zip, exdir = outdir)

# Reading GEDI level2B data (h5 file)
level2b <- readLevel2B(level2Bpath=level2Bpath)

close(level2b)
```


readLevel3

Description

Read GEDI Level 3 gridded metrics

Usage

```
readLevel3(x, metrics = NULL)
```

Arguments

x: One or more local paths, GDAL-readable URLs, or a 'SpatRaster'.

metrics: Optional layer names to retain.

Value

A 'terra::SpatRaster'.

See Also

<https://daac.ornl.gov/GEDI/guides/GEDI_L3_LandSurface_Metrics_V3.html>

readLevel4A

Description

Opens a GEDI04_A HDF5 granule containing footprint-level aboveground biomass density estimates.

Usage

```
readLevel4A(level4Apath)
```

Arguments

level4Apath: Local file path or Earthdata Cloud URL pointing to a GEDI04_A HDF5 granule.

Value

A 'gedi.level4a' object. Close it with 'close()'.

See Also

<https://daac.ornl.gov/GEDI/guides/GEDI_L4A_AGB_Density_V3.html>

readLevel4B

Description

Read GEDI Level 4B gridded biomass

Usage

```
readLevel4B(x, metrics = NULL)
```

Arguments

x: One or more local paths, GDAL-readable URLs, or a 'SpatRaster'.

metrics: Optional layer names to retain.

Value

A 'terra::SpatRaster'.

See Also

```
\Sexpr[results=rd]{{tools:::Rd_expr_doi("10.3334/ORNLDAAAC/2299")}}
```

resampleLevel3

Description

Resample a GEDI Level 3 raster

Usage

```
resampleLevel3(x, template, method = "bilinear", ...)
```

Arguments

x: A raster path or 'SpatRaster'.

template: Target 'SpatRaster'.

method: Resampling method.

...: Passed to 'terra::resample()'.

Value

A 'terra::SpatRaster'.

resampleLevel4B

Description

Resample a GEDI Level 4B raster

Usage

```
resampleLevel4B(x, template, method = "bilinear", ...)
```

Arguments

x: A raster path or 'SpatRaster'.

template: Target 'SpatRaster'.

method: Resampling method.

...: Passed to 'terra::resample()'.

Value

A 'terra::SpatRaster'.

rGEDI-package

Description

The rGEDI package provides functions for i) downloading, ii) visualizing, iii) clipping, iv) gridding, iv) simulating and v) exporting GEDI data.

Note

See more details about GEDI data in <<https://gedi.umd.edu/dataproducts/products/>>.

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See Also

For comprehensive examples refer to <<https://github.com/carlos-alberto-silva/rGEDI/blob/master/README.md>>

rGEDI_configure

Description

Configure optional rGEDI cloud and Earth Engine support

Usage

```
rGEDI_configure(  
    python = NULL,  
    install = FALSE,  
    earth_engine_project = Sys.getenv("EE_PROJECT", unset = ""),  
    quiet = FALSE  
)
```

Arguments

python: Optional Python executable or environment name.

install: Logical; install missing Python modules.

earth_engine_project: Optional Google Earth Engine project ID.

quiet: Logical.

Value

Invisibly returns a capability table.

sampleGEDI

Description

Sample GEDI footprint observations

Usage

```
sampleGEDI(x, method)
```

Arguments

x: A data frame or data table of GEDI footprints.

method: A specification returned by one of the sampling constructors.

Value

A `data.table::data.table`.

search_datasets

Description

Search the GEDI datasets available in Earth Engine

Usage

```
search_datasets(..., operator = c("and", "or"))
```

Arguments

...: Words matched against the product name, catalog ID, and title.

operator: Use "and" or "or" when several words are supplied.

Value

A data table containing product, catalog ID, version, and title.

slope

Description

Earth Engine terrain slope

Usage

```
slope(x)
```

Arguments

x: Earth Engine elevation image.

Value

An Earth Engine image.

spacedSampling

Description

GEDI minimum-distance sampling specification

Usage

```
spacedSampling(size, radius, lon = NULL, lat = NULL)
```

Arguments

size: Number or fraction of observations to select.

radius: Minimum separation in meters.

lon, lat: Optional coordinate column names.

stratifiedSampling

Description

GEDi attribute-stratified sampling specification

Usage

```
stratifiedSampling(size, variable, breaks = NULL)
```

Arguments

size: Number or fraction per stratum.

variable: Column used as strata, or a numeric column to bin.

breaks: Optional breaks for a numeric variable.

to_vect

Description

Convert GEDI footprints to a spatial vector

Usage

```
to_vect(x, lon = NULL, lat = NULL, crs = "EPSG:4326")
```

Arguments

x: GEDI footprint table.

lon, lat: Optional coordinate column names.

crs: Coordinate reference system.

Value

A 'terra::SpatVector'.

varSel

Description

Select predictor variables for a GEDI model

Usage

```
varSel(  
  x,  
  y,  
  method = c("correlation", "randomForest", "rfe"),  
  threshold = 0.1,  
  seed = NULL,  
  ...  
)
```

Arguments

x: Predictor data frame.

y: Response vector.

method: Selection method. "rfe" recursively removes the least important predictor and selects the smallest model whose out-of-bag error is within one standard error of the minimum.

threshold: Minimum absolute correlation or scaled Random Forest importance. For 'method = "rfe"', it is applied after model selection; use '0' to retain the RFE-selected subset.

seed: Optional random seed.

...: Passed to 'randomForest' when requested.

Value

An object of class 'gedi_var_selection'.

vect_as_ee

Description

Convert a vector object to an Earth Engine feature collection

Usage

```
vect_as_ee(x)
```

Arguments

x: 'sf', 'sfc', 'SpatVector', or vector file path.

Value

An Earth Engine 'FeatureCollection'.